

KISSsoft – Student license

File

Name: Unnamed
Changed by: robin on: 08.12.2022 at: 02:12:26

TwoStageGearBox

Power inputs:

_O.Input			
Speed	[1/min]	1500	
Torque	[Nm]	94.945	
Power	[kW]	14.914	

Power outputs:

_O.Output			
Speed	[1/min]	-587.838	
Torque	[Nm]	240.941	
Power	[kW]	14.832	

Inhalt

_O.DefaultGeometry.GearBox.GearPair_const12.GearPair_const12_calc_____	3
_O.DefaultGeometry.GearBox.GearPair_const34.GearPair_const34_calc_____	14
_O.DefaultGeometry.GearBox.Shaft1.Shaft1_calc_____	25
_O.DefaultGeometry.GearBox.Shaft2.Shaft2_calc_____	31
_O.DefaultGeometry.GearBox.Shaft3.Shaft3_calc_____	48

_O.DefaultGeometry.GearBox.GearPair_const12.GearPair_const12_calc

1 Overview

Calculation method	AGMA 2001-D04		
Drawing or article number:			
Gear 1:	z1(GearPair_const12)		
Gear 2:	z2(GearPair_const12)		
	----- Gear 1 ----- Gear 2 -----		
Transmitted power (kW, hp, ft*lb/s)	[P]	14.912 ,19.998 ,10999	
Speed (1/min)	[n]	1500.0	906.3
Number of load cycles (in mio.)	[NL]	1350.000	815.625
Torque (Nm, ft*lb)	[T]	94.9 ,70.0	
Torque (Nm, ft*lb)	[T]		157.1 ,115.9
Overload factor	[Ko]	1.25	
Required service life (h)	[H]	15000.00	
Gear driving (+) / driven (-)		+	-
Working flank Gear 1:	Right flank		
Gear 1 direction of rotation:	Clockwise		
Gearbox type:	Commercial enclosed gear units		

2 Tooth geometry

Geometry calculation according to	ISO 21771:2007		
	----- Gear 1 ----- Gear 2 -----		
Center distance (in, mm)	[a]	3.0992,78.720	
Center distance tolerance	ISO 286:2010 Measure js7		
Normal Diametral Pitch (1/in)	[Pnd]	12.70000	
Transverse Diametral Pitch (1/in)	[Pd]	12.42247	
Normal module (in, mm)	[mn]	0.07874,2.0000	
Normal pressure angle (°)	[αn]	22.5000	
Helix angle at reference circle (°)	[β]	12.0000	
Number of teeth	[z]	29	48
Facewidth (in)	[b]	1.100	1.000
Hand of gear		left	right
Accuracy grade	[Q-AGMA 2015-1-A01]	A7	A7
Inner diameter (in)	[di]	0.000	0.000
Inner diameter of gear rim (in)	[dbi]	0.000	0.000

3 Materials

Gear 1

Nitralloy N, Grade 2(AGMA), Nitriding steel, nitrided, AGMA2001: Surface 90.0HR15HN; Core 300HB

Gear 2

Nitralloy N, Grade 2(AGMA), Nitriding steel, nitrided, AGMA2001: Surface 90.0HR15HN; Core 300HB

		----- Gear 1 -----	Gear 2 -----
Surface hardness		HV 687	HV 687
Allowable bending stress number (lbf/in ²)	[sat]	50790.00	50790.00
Allowable bending stress number (N/mm ²)	[sat]	350.18	350.18
Allowable contact stress number (lbf/in ²)	[sac]	188000.00	188000.00
Allowable contact stress number (N/mm ²)	[sac]	1296.21	1296.21
Young's modulus (lbf/in ²)	[E]	30000000	30000000
Poisson's ratio	[ν]	0.300	0.300
Tensile strength (lbf/in ²)	[σB]	159542.18	159542.18
Yield point (lbf/in ²)	[σS]	130534.52	130534.52

3.1 Roughnesses

		----- Gear 1 -----	Gear 2 -----
Roughness average value DS, flank (mil)	[RAH]	0.094	0.094
Roughness average value DS, root (mil)	[RAF]	0.094	0.094
Mean roughness height, Rz, flank (mil)	[RZH]	0.630	0.630
Mean roughness height, Rz, root (mil)	[RZF]	0.630	0.630

3.2 Lubrication

Lubrication type	Oil bath lubrication		
Type of oil	ISO-VG 220		
Lubricant base	Mineral-oil base		
Oil nominal kinematic viscosity at 40°C (cSt)	[ν40]	220.00	
Oil nominal kinematic viscosity at 100°C (cSt)	[ν100]	17.50	
Specific density at 15°C (lb/ft ³)	[ρ]	55.874	
Oil temperature (°F)	[TS]	158.000	

4 Geometry

4.1 Reference profiles

Gear reference profile 1

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.401
Addendum coefficient	[haP*]	1.000
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Smallest radius of curvature, root rounding (in)	[ρmin.e/i]	0.0341/0.0343

Gear reference profile 2

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.401
Addendum coefficient	[haP*]	1.000
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Smallest radius of curvature, root rounding (in)	[ρmin.e/i]	0.0325/0.0326

4.1.1 Information on final machining

		----- Gear 1 -----	Gear 2 -----
Dedendum reference profile	[hfP*]	1.250	1.250
Tooth root radius Refer. profile	[ρfP*]	0.380	0.380
Addendum Reference profile	[haP*]	1.000	1.000
Protuberance height coefficient	[hprP*]	0.000	0.000
Protuberance angle (°)	[αprP]	0.000	0.000
Tip form height coefficient	[hFaP*]	0.000	0.000
Ramp angle (°)	[αKP]	0.000	0.000
Type of profile modification:	none (only running-in)		
Tip relief by running in (mil)	[Ca L/R]	0.11 /0.11	0.11 /0.11

4.2 Basic data

Overall transmission ratio	[itot]	-1.655	
Gear ratio	[u]	1.655	
Transverse module (in)	[mt]	0.0805	
Transverse pressure angle (°)	[αt]	22.951	
Working pressure angle (°)	[αwt]	22.951	
	[αwt.e/i]	22.977 /22.925	
Working pressure angle at normal section (°)	[αwn]	22.500	
Base helix angle (°)	[βb]	11.075	
Helix angle at operating pitch circle (°)	[βw]	12.000	
Reference center distance (in)	[ad]	3.0992	
Sum of profile shift coefficients	[Σxi]	-0.0000	
		----- Gear 1 -----	Gear 2 -----
Profile shift coefficient	[x]	-0.0000	-0.0000
Generating profile shift coefficient	[xE.e/i]	-0.0422/-0.0664	-0.0422/-0.0664
Virtual gear no. of teeth	[zn]	30.784	50.952
Involute length (in)	[l_dFa-l_dFf]	0.1557	0.1623

4.3 Diameters and their allowances

		----- Gear 1 -----	Gear 2 -----
Reference diameter (in)	[d]	2.3345	3.8640
Base diameter (in)	[db]	2.1497	3.5581
Tip alteration (in)	[k*mn]	0.0000	0.0000
Tip diameter (in)	[da]	2.4920	4.0214
(in)	[da.e/i]	2.4920/2.4908	4.0214/4.0201
Tip diameter allowances (in)	[Ada.e/i]	0.0000/-0.0012	0.0000/-0.0014

Tip form diameter (in)	[dFa]	2.4920	4.0214
(in)	[dFa.e/i]	2.4920/2.4908	4.0214/4.0201
Active tip diameter (in)	[dNa]	2.4920	4.0214
(in)	[dNa.e/i]	2.4920/2.4908	4.0214/4.0201
Operating pitch diameter (in)	[dw]	2.3345	3.8640
(in)	[dw.e/i]	2.3349/2.3340	3.8647/3.8632
Root diameter (in)	[df]	2.1376	3.6671
Generated root diameter with xE (in)	[df.e/i]	2.1310/2.1272	3.6605/3.6567
Active root diameter (in)	[dNf]	2.2172	3.7413
(in)	[dNf.e/i]	2.2187/2.2165	3.7430/3.7404
Root form diameter (in)	[dFf]	2.2071	3.7233
(in)	[dFf.e/i]	2.2033/2.2012	3.7183/3.7154

4.4 Tip clearances and tooth heights

		----- Gear 1 -----	Gear 2 -----
Theoretical tip clearance (in)	[c]	0.0197	0.0197
Effective tip clearance (in)	[c.e/i]	0.0261/0.0224	0.0262/0.0224
Reserve (dNf-dFf)/2 (in)	[cF.e/i]	0.0087/0.0066	0.0138/0.0111
Addendum, $m_n(h_{aP}^*+x+k)$ (in)	[ha]	0.0787	0.0787
(in)	[ha.e/i]	0.0787/0.0781	0.0787/0.0781
Dedendum, $m_n(h_{fP}^*-x)$ (in)	[hf]	0.0984	0.0984
(in)	[hf.e/i]	0.1018/0.1037	0.1018/0.1037
Tooth height (in)	[h]	0.1772	0.1772

4.5 Roll angle

		----- Gear 1 -----	Gear 2 -----
Roll angle at dFa (°)	[ξFa.e/i]	33.595 /33.533	30.178 /30.130
Roll angle to dNf (°)	[ξNf.e/i]	14.632 /14.392	18.711 /18.576
Roll angle at dFf (°)	[ξFf.e/i]	12.878 /12.618	17.385 /17.228

4.6 Tooth thickness and pitch

		----- Gear 1 -----	Gear 2 -----
Tooth thickness, arc, in module	[sn*]	1.5708	1.5708
Normal tooth thickness at tip circle (in)	[san]	0.0511	0.0538
(in)	[san.e/i]	0.0488/0.0465	0.0516/0.0493
Normal tooth thickness at tip form circle (in)	[sFan]	0.0511	0.0538
(in)	[sFan.e/i]	0.0488/0.0465	0.0516/0.0493
Normal space width at root circle (in)	[efn]	0.0000	0.0533
(in)	[efn.e/i]	0.0000/0.0000	0.0542/0.0547
Pitch on reference circle (in)	[pt]	0.2529	0.2529
Base pitch (in)	[pbt]	0.2329	0.2329
Transverse pitch on contact-path (in)	[pet]	0.2329	0.2329
Lead height (in)	[pz]	34.5037	57.1095
Axial pitch (in)	[px]	1.1898	1.1898

4.7 Sliding

		----- Gear 1 -----	Gear 2 -----
Max. sliding velocity at tip (ft/s)	[vga]	3.676	3.856
Specific sliding at the tip	[ζa]	0.446	0.520
Specific sliding at the root	[ζf]	-1.085	-0.804
Mean specific sliding	[ζm]	0.484	
Sliding factor on tip	[Kga]	0.241	0.252
Sliding factor on root	[Kgf]	-0.252	-0.241

4.8 Contact ratios

		----- Pair -----	
Minimal length of contact line (in)	[Lmin]	1.4808	
Transverse contact ratio	[εα]	1.540	
	[εα.e/m/i]	1.547 / 1.535/ 1.523	
Overlap ratio	[εβ]	0.840	
Total contact ratio	[εγ]	2.381	
	[εγ.e/m/i]	2.387 / 2.375/ 2.363	
Length of path of contact (in)	[ga]	0.3587	
(in)	[ga.e/i]	0.3602/ 0.3546	
		----- Gear 1 -----	Gear 2 -----
Addendum contact ratio	[ε]	0.752	0.789
	[ε.e/i]	0.754 / 0.744	0.793 / 0.778
Length T1-A and T2-A (in)	[T1A,T2A]	0.2715	0.9370
(in)	[.e/i]	0.2700/ 0.2745	0.9370/ 0.9355
Length T1-B and T2-B (in)	[T1B,T2B]	0.3974	0.8112
(in)	[.e/i]	0.3974/ 0.3962	0.8097/ 0.8139
Length T1-C and T2-C (in)	[T1C,T2C]	0.4552	0.7534
(in)	[.e/i]	0.4546/ 0.4557	0.7524/ 0.7543
Length T1-D and T2-D (in)	[T1D,T2D]	0.5044	0.7041
(in)	[.e/i]	0.5029/ 0.5074	0.7041/ 0.7027
Length T1-E and T2-E (in)	[T1E,T2E]	0.6302	0.5783
(in)	[.e/i]	0.6302/ 0.6291	0.5768/ 0.5810
Length T1-T2 (in)	[T1T2]	1.2085	
(in)	[.e/i]	1.2070/ 1.2100	
Diameter of single contact point B (in)	[d-B]	2.2919	3.9105
(in)	[d-B.e/i]	2.2919/ 2.2911	3.9092/ 3.9127
Diameter of single contact point D (in)	[d-D]	2.3746	3.8267
(in)	[d-D.e/i]	2.3733/ 2.3771	3.8267/ 3.8256

5 General influence factors

5.1 Forces and velocity

Calculated at the operating pitch circle:

Nominal circumferential force (lb)	[Wt,Ftw]	719.86	(3202.11 N)
Axial force (lb)	[Faw]	153.01	(680.63 N)
Radial force (lb)	[Frw]	304.84	(1355.99 N)
Net face width of narrowest member (in)	[F,b]	1.00	(25.40 mm)
Circumferential force per in (lb/in)	[w]	719.86	(126.07 N/mm)
Pitch line velocity (ft/min)	[vt]	916.75	(4.66 m/s)

5.2 General

Gearbox type:	Commercial enclosed gear units	
Mesh alignment factor	[Cma,KHma]	0.143
Mounting procedure:	Contact improved by adjusting at assembly	
Mesh alignment correction factor	[Ce,KHe]	0.800
Gearing:	crowned	
Lead correction factor	[Cmc,KHmc]	0.800
Pinion proportion factor	[Cpf,KHpf]	0.025
Small offset	[s1/s < 0.175]	
Pinion proportion modifier	[Cpm,KHpm]	1.000
Face load distribution factor	[Cmf,KHβ]	1.111
Load distribution factor	[Km,KHβ]	1.111
Dynamic factor	[Kv]	1.189
Transmission accuracy level number	[Av]	8

6 Calculation of tooth root strength (fracture)

		----- Gear 1 -----	Gear 2 -----
Rim thickness factor	[KB]	1.00	1.00
Size factor	[KS]	1.00	1.00
Limiting Variation in action (in/10000)	[LimVarAc]	2.00	
Load sharing:			
0 = No, Loaded at tip, 1 = Yes, Loaded at HPSTC		1	1
Calc. as helical gear (0) / as LACR (1)		1	1
Load angle (°)	[φnL]	21.78	22.34
Calculation of factor Y following	AGMA 908-B89		
Calculated with generating profile shift coefficient	[xE.m]	-0.0543	-0.0543
		(in) (mm)	
Heigth of Lewis parabola	[hF]	0.082 , 2.07	0.085 , 2.15
Tooth thickness at critical section	[sF]	0.167 , 4.24	0.176 , 4.47
Radius at curvature of fillet curve	[ρF]	0.034 , 0.86	0.032 , 0.82
Helical factor	[Ch]	1.00	
Helix angle factor	[Kψ]	1.00	
Tooth form factor Y	[Y]	0.832	0.902
Stress correction factor	[Kf]	1.894	1.930
Helical overlap factor	[Cψ]	1.25	
Load sharing ratio	[mN]	1.00	
Bending strength geometry factor J	[J,YJ]	0.549	0.584
Bending stress number (lb/in²)	[st]	26883.14	25271.47
Bending stress number (N/mm²)	[σF]	185.35	174.24
Stress cycle factor	[YN]	0.933	0.941
for general applications			
Allowable bending stress number (lb/in²)	[sat]	50790.00	50790.00
Allowable bending stress number (N/mm²)	[σFP]	350.18	350.18
Temperature factor	[KT,Yθ]	1.000	1.000
Reliability factor (99.00 %)	[KR,YZ]	1.000	

Reverse loading factor	[-]	1.000	1.000
Effective allow. b.s.n. (lb/in ²)	[sateff]	47364.58	47791.33
Effective allow. b.s.n. (N/mm ²)	[σFPeff]	326.57	329.51

Allowable transmitted power (hp,kW) [Pat] 35.23 ,26.27 37.82 ,28.20

Note: Allowable transmitted power P_{at} calculated including the following values: K_o , K_R , S_{Fmin}

Allowable transmitted power at unity service factor (hp,kW) [Patu] 44.04 ,32.84 47.27 ,35.25

Note: Allowable transmitted power P_{atu} calculated with $K_o=1$, $K_R=1$, $S_{Fmin}=1$

Unit load (lb/in ²)	[UL]	9142.262
Unit load (N/mm ²)	[UL]	63.034

Allowable unit load (lb/in ²)	[Uat]	16107.470	17289.097
Allowable unit load (N/mm ²)	[Uat]	111.057	119.204
Required safety factor	[SFmin]	1.00	1.00
Safety factor, Bending	[sateff/st]	1.76	1.89

Note: Materials with HB > 400: Yield strength not checked.

7 Calculation of flank strength (pitting)

		----- Gear 1 -----	Gear 2 -----
Elastic coefficient (√lb/in)	[Cp]	2290.0	
Elastic coefficient (√N/mm)	[ZE]	190.20	
Size factor	[Ks]	1.000	1.000
Load sharing ratio	[mN]	1.000	
Helical overlap factor	[Cψ]	1.250	
Geometry factor	[I,ZI]	0.164	
Contact stress number (lb/in ²)	[sc]	127461	
Contact stress number (N/mm ²)	[σH]	878.81	
Stress cycle factor	[ZN]	0.893	0.904
(for general applications)			
Surface condition factor	[Cf,ZR]	1.000	1.000
Hardness ratio factor	[CH,ZW]	1.000	1.000
Temperature factor	[KT,Yθ]	1.00	1.00
Reliability factor (99.00 %)	[KR,YZ]	1.00	
Allowable contact stress number (lb/in ²)	[sac]	188000.00	188000.00
Allowable contact stress number (N/mm ²)	[σHP]	1296.21	1296.21
Effective allow. c.s.n. (lb/in ²)	[saceff]	167945.52	169903.30
Effective allow. c.s.n. (N/mm ²)	[σHPeff]	1157.94	1171.44

Allowable transmitted power (hp,kW) [Pac] 34.72 ,25.89 kW 35.53 ,26.50 kW

Note: Allowable transmitted power P_{ac} calculated including the following values: K_o , K_R , S_{Hmin}

Allowable transmitted power at unity service factors (hp,kW) [Pacu] 43.40 ,32.36 kW 44.42 ,33.12 kW

Note: Allowable transmitted power P_{acu} calculated including the following values: $K_o=1$, $K_R=1$, $S_{Hmin}=1$

Contact load factor (lb/in ²)	[K]	494.7	(3.411 N/mm ²)
Allowable contact load factor (lb/in ²)	[Kac]	858.81	878.95
Allowable contact load factor (N/mm ²)	[Kac]	5.92	6.06
Note: K_{ac} calculated with CG = (0.623, $S_{Hmin}=1$)			
Required safety factor	[SHmin]	1.00	1.00
Safety factor, Pitting	[saceff/sc]	1.32	1.33

8 Service factors

Service factor for tooth root	[KSF]	2.20	2.36
Service factor for pitting	[CSF]	2.17	2.22
Service factor for gear set	[SF=min(KSF,CSF)]	2.17	

Note: Service factors calculated with $K_o=1$, $K_R=1$, $S_{Fmin}=1$, $S_{Hmin}=1$. The service factor for pitting (C_{SF}) is a function of the power while the safety factor (S_H) is function of the stress. S_H is comparable with $\sqrt{C_{SF}}$.

The equations used for P_{at} , P_{ac} , P_{acu} , P_{atu} , P_a are according to AGMA 2001.

Transmittable power including required service factors K_{SFmin} , C_{SFmin} (hp)	43.40 (32.36 kW)
$K_{SFmin}=1.00$, $C_{SFmin}=1.00$	

9 Measurements for tooth thickness

9.1 Tooth thickness tolerances

		----- Gear 1 -----	Gear 2 -----
Tooth thickness tolerance		DIN 3967 cd25	DIN 3967 cd25
Tooth thickness allowance (normal section) (in)	[As.e/i]	-0.0028/-0.0043	-0.0028/-0.0043

9.2 Base tangent lengths

		----- Gear 1 -----	Gear 2 -----
Number of teeth spanned	[k]	4.000	7.000
Base tangent length (no backlash) (in)	[Wk]	0.8482	1.5655
Base tangent length with allowance (in)	[Wk.e/i]	0.8456/0.8442	1.5629/1.5615
(in)	[ΔWk.e/i]	-0.0025/-0.0040	-0.0025/-0.0040
Diameter of measuring circle (in)	[dMWk.m]	2.3041	3.8743

9.3 Measurement over balls and pins

		----- Gear 1 -----	Gear 2 -----
Theoretical diameter of ball/pin (in)	[DM]	0.1368	0.1356
Effective diameter of ball/pin (in)	[DMeff]	0.1440	0.1440
Radial single-ball measurement backlash free (in)	[MrK]	1.2752	2.0410
Radial single-ball measurement (in)	[MrK.e/i]	1.2723/1.2707	2.0379/2.0362
Diameter of measuring circle (in)	[dMMr.m]	2.3397	3.8722
Diametral measurement over two balls without clearance (in)	[MdK]	2.5469	4.0819
Diametral measurement over two balls (in)	[MdK.e/i]	2.5411/2.5378	4.0758/4.0723
Diametral measurement over pins without clearance (in)	[MdR]	2.5505	4.0819
Measurement over pins according to DIN 3960 (in)	[MdR.e/i]	2.5447/2.5413	4.0758/4.0723
Measurement over 2 pins, free, AGMA 2002 (in)	[dk2f.e/i]	2.5410/2.5376	0.0000/0.0000
Measurement over 2 pins, transverse, AGMA 2002 (in)	[dk2t.e/i]	2.5481/2.5448	0.0000/0.0000
Measurement over 3 pins, axial, AGMA 2002 (in)	[dk3A.e/i]	2.5447/2.5413	4.0758/4.0723

9.4 Tooth thickness

		----- Gear 1 -----	Gear 2 -----
Reference chordal height from da.m (in)	[ha]	0.0800	0.0793
Chordal tooth thickness (in)	[sc]	0.1236	0.1237
(in)	[sc.e/i]	0.1209/0.1194	0.1209/0.1194
Tooth thickness, arc (in)	[sn]	0.1237	0.1237
(in)	[sn.e/i]	0.1209/0.1194	0.1209/0.1194

9.5 Backlash

----- Pair -----		
Circumferential backlash from Aa (in)	[jtw_Aa.e/i]	0.0005/-0.0005
Radial backlash (in)	[jrw.e/i]	0.0111/ 0.0061
Circumferential backlash (transverse section) (in)	[jtw.e/i]	0.0094/0.0051
Normal backlash (in)	[jn.e/i]	0.0085/0.0046
Center distance allowances (in)	[Aa.e/i]	0.0006/-0.0006
Backlash free center distance (in)	[aControl.e/i]	3.0925/3.0887
Backlash free center distance, allowances (in)	[jta]	-0.0067/-0.0106
----- Gear 1 ----- Gear 2 -----		
dNf.i with aControl (in)	[dNf0.i]	2.2044 3.7248
Reserve (dNf0.i-dFf.e)/2 (in)	[cF0.i]	0.0006 0.0033
Tip clearance (in)	[c0.i(aControl)]	0.0125 0.0125
Torsional angle on input with output fixed:		
Total torsional angle (°)	[j.tSys]	0.4592/0.2520

10 Toothing tolerances

----- Gear 1 ----- Gear 2 -----			
Following AGMA 2000-A88			
Accuracy grade	[Q-AGMA2000]	10	10
Pitch Variation Allowable (mil)	[VpA]	0.343	0.374
Runout Radial Tolerance (mil)	[VrT]	1.063	1.220
Profile Tolerance (mil)	[VphiT]	0.433	0.472
Tooth Alignment Tolerance (mil)	[VpsiT]	0.346	0.323
Composite Tolerance, Tooth-to-Tooth (mil)	[VqT]	0.709	0.709
Composite Tolerance, Total (mil)	[VcqT]	1.772	1.929
AGMA <-> ISO: VpA <-> fpbT, VrT <-> FrT, VpsiT <-> FbT, VqT <-> fidT, VcqT <-> FidT			
According to AGMA 2015-1-A01 & 2015-2-A06			
Accuracy grade	[Q-AGMA2015]	A7	A7
Single pitch deviation (mil)	[fpT]	0.472	0.472
Total cumulative pitch deviation (mil)	[FpT]	1.772	1.850
Profile form deviation (mil)	[ffaT]	0.433	0.433
Profile slope deviation (mil)	[fHaT]	0.354	0.374
Total profile deviation (mil)	[FaT]	0.551	0.591
Helix form deviation (mil)	[ffβT]	0.472	0.472
Helix slope deviation (mil)	[fHβT]	0.472	0.472
Total helix deviation (mil)	[FβT]	0.669	0.669
Single flank composite, total (mil)	[FisT]	1.929	2.047
Single flank composite, tooth-to-tooth (mil)	[fisT]	0.177	0.185
Radial composite, total (mil)	[FidT]	1.811	1.890
Radial composite, tooth-to-tooth (mil)	[fidT]	0.335	0.354
According to AGMA 2015-2-B15			
Radial composite, total (mil)	[FidT]	2.835(R25)	2.87 (R25)
Radial composite, tooth-to-tooth (mil)	[fidT]	1.535(R25)	1.42 (R25)
Axis alignment tolerances recommendation acc. to ISO TR 10064-3:1996, Quality 7			
Maximum value for deviation error of axis (mil)	[fΣβ]	0.823	(Fβ=19.00)
Maximum value for inclination error of axes (mil)	[fΣδ]	1.646	

11 Modifications and determination of the tooth form

11.1 Data for the tooth form calculation

Data not available.

Please run the calculation in the "Tooth form" tab and open the main report again.

12 Supplementary data

Maximal possible center distance (eps_a=1.0)	[aMAX]	80.021
--	--------	--------

12.1 Masses, stiffnesses and moments of inertia

		----- Gear 1 -----	Gear 2 -----
Mass (lb)	[m]	1.309	3.283
Total mass (lb)	[mGes]	4.593	
Moment of inertia for system, relative to the input: calculation without consideration of the exact tooth shape			
Single gears, (da+df)/2...di (lb*ft²)	[J]	0.006091	0.04212
System (da+df)/2...di (lb*ft²)	[J]	0.02147	
Singular tooth stiffness (lbf/in/μin)	[c]	2.035	
Meshing stiffness (lbf/in/μin)	[cy]	2.859	

12.2 Wear, power loss, sound pressure level

Torsional stiffness at driving gear with fixed driven gear:		
Torsional stiffness (1E6*ft*lb/rad)	[cr]	0.234
Torsion when subjected to nominal torque (°)	[δcr]	0.017
Average coefficient of friction according to Niemann	[μm]	0.092
Wear sliding coef. by Niemann	[ζw]	0.745
Loss factor	[HV]	0.115
Gear power loss (hp)	[PVZ]	0.210
Meshing efficiency (%)	[ηz]	98.950
Sound pressure level based on Masuda, without PPTE/δs	[dB(A)]	63.808

12.3 Indications for the manufacturing by wire cutting

		----- Gear 1 -----	Gear 2 -----
Deviation from theoretical tooth trace (mil)	[WireErr]	4.84	2.92
Permissible deviation (mil)	[Fb/2]	0.33	0.33

13 Service life, damage

Required safety for tooth root	[S _{Fmin}]	1.00
Required safety for tooth flank	[S _{Hmin}]	1.00
Required service life	[H]	15000.00

Service life (calculated with required safeties):

System service life (h)	[H _{att}]	> 1000000
-------------------------	---------------------	-----------

		----- Gear 1 -----	Gear 2 -----
Tooth root service life (h)	[H _{Fatt}]	1e+06	1e+06
Tooth flank service life (h)	[H _{Hatt}]	1e+06	1e+06

Note: The entry 1e+006 h means that the Service life > 1,000,000 h.

13.1 Damage

Damage relative to the required service life (H, 15000.0 h)

F ₁ (%)	F ₂ (%)	H ₁ (%)	H ₂ (%)
0.0000	0.0000	0.0006	0.0004

14 Remarks

14.1 Conventions

- Specifications with **.e/i** mean: Maximum value **.e** and Minimum value **.i**, taking all tolerances into account.
- Specifications with **.m** mean: Mean value within tolerance.
- The center distance tolerances and the tooth thickness allowance are taken into account for the backlash tolerance. The maximum and minimum clearance corresponding to the largest and smallest allowances are shown. The calculation is performed for the operating pitch circle.
- Symbols used in **□**: [xx,yy] xx as used in AGMA 2001-D04, yy as used in AGMA 2101-D04

14.2 Calculations and factors

- $S_{ateff} = S_{at} \cdot K_L / K_T / K_R \cdot K_{wb} / S_F$ ($S_F = 1.0$)
- LACR = Spur gear or helical gear with $\epsilon_\beta (m_F) < 1.0$
- PSTC = Point of Single Tooth Contact
- Calculation of Z_β acc. to Corrigendum 1 ISO 6336-2:2008 with $Z_\beta = 1 / (\cos(\beta))^{0.5}$

_O.DefaultGeometry.GearBox.GearPair_const34.GearPair_const34_calc

1 Overview

Calculation method	AGMA 2001-D04		
Drawing or article number:			
Gear 1:	z3(GearPair_const34)		
Gear 2:	z4(GearPair_const34)		
	----- Gear 1 ----- Gear 2 -----		
Transmitted power (kW, hp, ft*lb/s)	[P]	14.833 ,19.892 ,10941	
Speed (1/min)	[n]	906.3	587.8
Number of load cycles (in mio.)	[NL]	815.625	529.054
Torque (Nm, ft*lb)	[T]	156.3 ,115.3	
Torque (Nm, ft*lb)	[T]		241.0 ,177.7
Overload factor	[Ko]	1.25	
Required service life (h)	[H]	15000.00	
Gear driving (+) / driven (-)		+	-
Working flank Gear 1:	Right flank		
Gear 1 direction of rotation:	Clockwise		
Gearbox type:	Commercial enclosed gear units		

2 Tooth geometry

Geometry calculation according to	ISO 21771:2007		
	----- Gear 1 ----- Gear 2 -----		
Center distance (in, mm)	[a]	3.0991 ,78.717	
Center distance tolerance	ISO 286:2010 Measure js7		
Normal Diametral Pitch (1/in)	[Pnd]	10.15000	
Transverse Diametral Pitch (1/in)	[Pd]	9.92820	
Normal module (in, mm)	[mn]	0.09852,2.5025	
Normal pressure angle (°)	[αn]	22.5000	
Helix angle at reference circle (°)	[β]	12.0000	
Number of teeth	[z]	24	37
Facewidth (in)	[b]	1.390	1.380
Hand of gear		left	right
Accuracy grade	[Q-AGMA 2015-1-A01]	A7	A7
Inner diameter (in)	[di]	0.000	0.000
Inner diameter of gear rim (in)	[dbi]	0.000	0.000

3 Materials

Gear 1

Nitralloy N, Grade 2(AGMA), Nitriding steel, nitrided, AGMA2001: Surface 90.0HR15HN; Core 300HB

Gear 2

Nitralloy N, Grade 2(AGMA), Nitriding steel, nitrided, AGMA2001: Surface 90.0HR15HN; Core 300HB

		----- Gear 1 -----	Gear 2 -----
Surface hardness		HV 687	HV 687
Allowable bending stress number (lbf/in ²)	[sat]	50790.00	50790.00
Allowable bending stress number (N/mm ²)	[sat]	350.18	350.18
Allowable contact stress number (lbf/in ²)	[sac]	188000.00	188000.00
Allowable contact stress number (N/mm ²)	[sac]	1296.21	1296.21
Young's modulus (lbf/in ²)	[E]	30000000	30000000
Poisson's ratio	[ν]	0.300	0.300
Tensile strength (lbf/in ²)	[σB]	159542.18	159542.18
Yield point (lbf/in ²)	[σS]	130534.52	130534.52

3.1 Roughnesses

		----- Gear 1 -----	Gear 2 -----
Roughness average value DS, flank (mil)	[RAH]	0.094	0.094
Roughness average value DS, root (mil)	[RAF]	0.094	0.094
Mean roughness height, Rz, flank (mil)	[RZH]	0.630	0.630
Mean roughness height, Rz, root (mil)	[RZF]	0.630	0.630

3.2 Lubrication

Lubrication type	Oil bath lubrication		
Type of oil	ISO-VG 220		
Lubricant base	Mineral-oil base		
Oil nominal kinematic viscosity at 40°C (cSt)	[ν40]	220.00	
Oil nominal kinematic viscosity at 100°C (cSt)	[ν100]	17.50	
Specific density at 15°C (lb/ft ³)	[ρ]	55.874	
Oil temperature (°F)	[TS]	158.000	

4 Geometry

4.1 Reference profiles

Gear reference profile 1

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.401
Addendum coefficient	[haP*]	1.000
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Smallest radius of curvature, root rounding (in)	[ρmin.e/i]	0.0410/0.0412

Gear reference profile 2

Reference profile	1.25 / 0.38 / 1.0 ISO 53:1998 Profil A	
Dedendum coefficient	[hfP*]	1.250
Root radius factor	[ρfP*]	0.380
	[ρfPmax*]	0.401
Addendum coefficient	[haP*]	1.000
Tip radius factor	[ρaP*]	0.000
Protuberance height coefficient	[hprP*]	0.000
Protuberance angle	[αprP]	0.000
Tip form height coefficient	[hFaP*]	0.000
Ramp angle	[αKP]	0.000
	not topping	
Smallest radius of curvature, root rounding (in)	[ρmin.e/i]	0.0410/0.0411

4.1.1 Information on final machining

		----- Gear 1 -----	Gear 2 -----
Dedendum reference profile	[hfP*]	1.250	1.250
Tooth root radius Refer. profile	[ρfP*]	0.380	0.380
Addendum Reference profile	[haP*]	1.000	1.000
Protuberance height coefficient	[hprP*]	0.000	0.000
Protuberance angle (°)	[αprP]	0.000	0.000
Tip form height coefficient	[hFaP*]	0.000	0.000
Ramp angle (°)	[αKP]	0.000	0.000
Type of profile modification:	none (only running-in)		
Tip relief by running in (mil)	[Ca L/R]	0.11 /0.11	0.11 /0.11

4.2 Basic data

Overall transmission ratio	[itot]	-1.542	
Gear ratio	[u]	1.542	
Transverse module (in)	[mt]	0.1007	
Transverse pressure angle (°)	[αt]	22.951	
Working pressure angle (°)	[αwt]	24.104	
(°)	[αwt.e/i]	24.129 /24.080	
Working pressure angle at normal section (°)	[αwn]	23.628	
Base helix angle (°)	[βb]	11.075	
Helix angle at operating pitch circle (°)	[βw]	12.103	
Reference center distance (in)	[ad]	3.0721	
Sum of profile shift coefficients	[Σxi]	0.2811	
		----- Gear 1 -----	Gear 2 -----
Profile shift coefficient	[x]	0.2169	0.0642
Generating profile shift coefficient	[xE.e/i]	0.1831/0.1638	0.0304/0.0111
Virtual gear no. of teeth	[zn]	25.476	39.276
Involute length (in)	[l_dFa-l_dFf]	0.2031	0.2013

4.3 Diameters and their allowances

		----- Gear 1 -----	Gear 2 -----
Reference diameter (in)	[d]	2.4174	3.7268
Base diameter (in)	[db]	2.2260	3.4317
Tip alteration (in)	[k*mn]	-0.0006	-0.0006
Tip diameter (in)	[da]	2.6559	3.9352
(in)	[da.e/i]	2.6559/2.6547	3.9352/3.9338
Tip diameter allowances (in)	[Ada.e/i]	0.0000/-0.0012	0.0000/-0.0014

Tip form diameter (in)	[dFa]	2.6559	3.9352
(in)	[dFa.e/i]	2.6559/2.6547	3.9352/3.9338
Active tip diameter (in)	[dNa]	2.6559	3.9352
(in)	[dNa.e/i]	2.6559/2.6547	3.9352/3.9338
Operating pitch diameter (in)	[dw]	2.4386	3.7596
(in)	[dw.e/i]	2.4391/2.4382	3.7603/3.7588
Root diameter (in)	[df]	2.2138	3.4931
Generated root diameter with xE (in)	[df.e/i]	2.2071/2.2033	3.4864/3.4826
Active root diameter (in)	[dNf]	2.3069	3.5985
(in)	[dNf.e/i]	2.3084/2.3061	3.6000/3.5976
Root form diameter (in)	[dFf]	2.2904	3.5669
(in)	[dFf.e/i]	2.2864/2.2842	3.5623/3.5597

4.4 Tip clearances and tooth heights

		----- Gear 1 -----	Gear 2 -----
Theoretical tip clearance (in)	[c]	0.0246	0.0246
Effective tip clearance (in)	[c.e/i]	0.0310/0.0273	0.0311/0.0273
Reserve (dNf-dFf)/2 (in)	[cF.e/i]	0.0121/0.0099	0.0202/0.0177
Addendum, $m_n(h_{aP}^*+x+k)$ (in)	[ha]	0.1193	0.1042
(in)	[ha.e/i]	0.1193/0.1187	0.1042/0.1035
Dedendum, $m_n(h_{fP}^*-x)$ (in)	[hf]	0.1018	0.1168
(in)	[hf.e/i]	0.1051/0.1070	0.1202/0.1221
Tooth height (in)	[h]	0.2210	0.2210

4.5 Roll angle

		----- Gear 1 -----	Gear 2 -----
Roll angle at dFa (°)	[ξFa.e/i]	37.288 /37.232	32.153 /32.106
Roll angle to dNf (°)	[ξNf.e/i]	15.733 /15.511	18.161 /18.028
Roll angle at dFf (°)	[ξFf.e/i]	13.438 /13.187	15.953 /15.790

4.6 Tooth thickness and pitch

		----- Gear 1 -----	Gear 2 -----
Tooth thickness, arc, in module	[sn*]	1.7505	1.6239
Normal tooth thickness at tip circle (in)	[san]	0.0568	0.0652
(in)	[san.e/i]	0.0545/0.0521	0.0631/0.0607
Normal tooth thickness at tip form circle (in)	[sFan]	0.0568	0.0652
(in)	[sFan.e/i]	0.0545/0.0521	0.0631/0.0607
Normal space width at root circle (in)	[efn]	0.0000	0.0697
(in)	[efn.e/i]	0.0000/0.0000	0.0710/0.0717
Pitch on reference circle (in)	[pt]	0.3164	0.3164
Base pitch (in)	[pbt]	0.2914	0.2914
Transverse pitch on contact-path (in)	[pet]	0.2914	0.2914
Lead height (in)	[pz]	35.7286	55.0816
Axial pitch (in)	[px]	1.4887	1.4887

4.7 Sliding

		----- Gear 1 -----	Gear 2 -----
Max. sliding velocity at tip (ft/s)	[vga]	2.951	2.545
Specific sliding at the tip	[ζa]	0.515	0.515
Specific sliding at the root	[ζf]	-1.063	-1.063
Mean specific sliding	[ζm]		0.515
Sliding factor on tip	[Kga]	0.306	0.264
Sliding factor on root	[Kgf]	-0.264	-0.306

4.8 Contact ratios

		----- Pair -----	
Minimal length of contact line (in)	[Lmin]	1.9732	
Transverse contact ratio	[εα]	1.447	
	[εα.e/m/i]	1.452 / 1.443 / 1.433	
Overlap ratio	[εβ]	0.927	
Total contact ratio	[εγ]	2.374	
	[εγ.e/m/i]	2.379 / 2.370 / 2.360	
Length of path of contact (in)	[ga]	0.4216	
(in)	[ga.e/i]	0.4230 / 0.4176	
		----- Gear 1 -----	Gear 2 -----
Addendum contact ratio	[ε]	0.777	0.670
	[ε.e/i]	0.779 / 0.771	0.673 / 0.662
Length T1-A and T2-A (in)	[T1A,T2A]	0.3028	0.9629
(in)	[.e/i]	0.3013 / 0.3056	0.9629 / 0.9615
Length T1-B and T2-B (in)	[T1B,T2B]	0.4330	0.8327
(in)	[.e/i]	0.4330 / 0.4319	0.8313 / 0.8352
Length T1-C and T2-C (in)	[T1C,T2C]	0.4980	0.7677
(in)	[.e/i]	0.4974 / 0.4985	0.7668 / 0.7686
Length T1-D and T2-D (in)	[T1D,T2D]	0.5941	0.6715
(in)	[.e/i]	0.5927 / 0.5970	0.6715 / 0.6701
Length T1-E and T2-E (in)	[T1E,T2E]	0.7243	0.5413
(in)	[.e/i]	0.7243 / 0.7233	0.5399 / 0.5439
Length T1-T2 (in)	[T1T2]	1.2657	
(in)	[.e/i]	1.2642 / 1.2671	
Diameter of single contact point B (in)	[d-B]	2.3885	3.8145
(in)	[d-B.e/i]	2.3885 / 2.3877	3.8133 / 3.8167
Diameter of single contact point D (in)	[d-D]	2.5233	3.6852
(in)	[d-D.e/i]	2.5219 / 2.5260	3.6852 / 3.6842

5 General influence factors

5.1 Forces and velocity

Calculated at the operating pitch circle:

Nominal circumferential force (lb)	[Wt,Ftw]	1134.56	(5046.77 N)
Axial force (lb)	[Faw]	243.28	(1082.17 N)
Radial force (lb)	[Frw]	507.62	(2257.99 N)
Net face width of narrowest member (in)	[F,b]	1.38	(35.05 mm)
Circumferential force per in (lb/in)	[w]	822.14	(143.98 N/mm)
Pitch line velocity (ft/min)	[vt]	578.58	(2.94 m/s)

5.2 General

Gearbox type:	Commercial enclosed gear units	
Mesh alignment factor	[Cma,KHma]	0.149
Mounting procedure:	Contact improved by adjusting at assembly	
Mesh alignment correction factor	[Ce,KHe]	0.800
Gearing:	crowned	
Lead correction factor	[Cmc,KHmc]	0.800
Pinion proportion factor	[Cpf,KHpf]	0.036
Small offset	[s1/s < 0.175]	
Pinion proportion modifier	[Cpm,KHpm]	1.000
Face load distribution factor	[Cmf,KHβ]	1.124
Load distribution factor	[Km,KHβ]	1.124
Dynamic factor	[Kv]	1.105
Transmission accuracy level number	[Av]	7

6 Calculation of tooth root strength (fracture)

		----- Gear 1 -----	Gear 2 -----
Rim thickness factor	[KB]	1.00	1.00
Size factor	[KS]	1.00	1.00
Limiting Variation in action (in/10000)	[LimVarAc]	5.00	
Load sharing:			
0 = No, Loaded at tip, 1 = Yes, Loaded at HPSTC		1	1
Calc. as helical gear (0) / as LACR (1)		1	1
Load angle (°)	[φnL]	24.28	23.30
Calculation of factor Y following	AGMA 908-B89		
Calculated with generating profile shift coefficient	[xE.m]	0.1735	0.0207
	(in) (mm)		
Height of Lewis parabola	[hF]	0.107 , 2.73	0.113 , 2.86
Tooth thickness at critical section	[sF]	0.214 , 5.43	0.216 , 5.49
Radius at curvature of fillet curve	[ρF]	0.041 , 1.04	0.041 , 1.04
Helical factor	[Ch]	1.00	
Helix angle factor	[Kψ]	1.00	
Tooth form factor Y	[Y]	0.853	0.812
Stress correction factor	[Kf]	1.886	1.858
Helical overlap factor	[Cψ]	1.24	
Load sharing ratio	[mN]	1.00	
Bending strength geometry factor J	[J,YJ]	0.562	0.543
Bending stress number (lb/in²)	[st]	22559.35	23329.74
Bending stress number (N/mm²)	[σF]	155.54	160.85
Stress cycle factor	[YN]	0.941	0.948
for general applications			
Allowable bending stress number (lb/in²)	[sat]	50790.00	50790.00
Allowable bending stress number (N/mm²)	[σFP]	350.18	350.18
Temperature factor	[KT,Yθ]	1.000	1.000
Reliability factor (99.00 %)	[KR,YZ]	1.000	

Reverse loading factor	[-]	1.000	1.000
Effective allow. b.s.n. (lb/in ²)	[sateff]	47791.33	48160.98
Effective allow. b.s.n. (N/mm ²)	[σFPeff]	329.51	332.06

Allowable transmitted power (hp,kW) [Pat] 42.14 ,31.42 41.06 ,30.62

Note: Allowable transmitted power P_{at} calculated including the following values: K_o , K_R , S_{Fmin}

Allowable transmitted power at unity service factor (hp,kW) [Patu] 52.68 ,39.28 51.33 ,38.28

Note: Allowable transmitted power P_{atu} calculated with $K_o=1$, $K_R=1$, $S_{Fmin}=1$

Unit load (lb/in ²)	[UL]	8344.770
Unit load (N/mm ²)	[UL]	57.535

Allowable unit load (lb/in ²)	[Uat]	17678.155	17226.606
Allowable unit load (N/mm ²)	[Uat]	121.887	118.773
Required safety factor	[SFmin]	1.00	1.00
Safety factor, Bending	[sateff/st]	2.12	2.06

Note: Materials with HB > 400: Yield strength not checked.

7 Calculation of flank strength (pitting)

		----- Gear 1 -----	Gear 2 -----
Elastic coefficient (√lb/in)	[Cp]	2290.0	
Elastic coefficient (√N/mm)	[ZE]	190.20	
Size factor	[Ks]	1.000	1.000
Load sharing ratio	[mN]	1.000	
Helical overlap factor	[Cψ]	1.243	
Geometry factor	[I,ZI]	0.165	
Contact stress number (lb/in ²)	[sc]	129136	
Contact stress number (N/mm ²)	[σH]	890.36	
Stress cycle factor (for general applications)	[ZN]	0.904	0.913
Surface condition factor	[Cf,ZR]	1.000	1.000
Hardness ratio factor	[CH,ZW]	1.000	1.000
Temperature factor	[KT,Yθ]	1.00	1.00
Reliability factor (99.00 %)	[KR,YZ]	1.00	
Allowable contact stress number (lb/in ²)	[sac]	188000.00	188000.00
Allowable contact stress number (N/mm ²)	[σHP]	1296.21	1296.21
Effective allow. c.s.n. (lb/in ²)	[saceff]	169903.30	171603.28
Effective allow. c.s.n. (N/mm ²)	[σHPeff]	1171.44	1183.16
Allowable transmitted power (hp,kW)	[Pac]	34.43 ,25.68 kW	35.13 ,26.19 kW
Note: Allowable transmitted power P_{ac} calculated including the following values: K_o , K_R , S_{Hmin}			
Allowable transmitted power at unity service factors (hp,kW)	[Pacu]	43.04 ,32.10 kW	43.91 ,32.74 kW
Note: Allowable transmitted power P_{acu} calculated including the following values: $K_o=1$, $K_R=1$, $S_{Hmin}=1$			
Contact load factor (lb/in ²)	[K]	555.8	(3.832 N/mm ²)
Allowable contact load factor (lb/in ²)	[Kac]	962.14	981.49
Allowable contact load factor (N/mm ²)	[Kac]	6.63	6.77
Note: K_{ac} calculated with $CG = (0.607, S_{Hmin}=1)$			
Required safety factor	[SHmin]	1.00	1.00
Safety factor, Pitting	[saceff/sc]	1.32	1.33

8 Service factors

Service factor for tooth root	[KSF]	2.65	2.58
Service factor for pitting	[CSF]	2.16	2.21

Service factor for gear set [SF=min(KSF,CSF)] 2.16

Note: Service factors calculated with $K_O=1$, $K_R=1$, $S_{Fmin}=1$, $S_{Hmin}=1$. The service factor for pitting (C_{SF}) is a function of the power while the safety factor (S_H) is function of the stress. S_H is comparable with $\sqrt{C_{SF}}$.

The equations used for P_{at} , P_{ac} , P_{acu} , P_{atu} , P_a are according to AGMA 2001.

Transmittable power including required service factors K_{SFmin} , C_{SFmin} (hp) 43.04 (32.10 kW)
 $K_{SFmin}=1.00$, $C_{SFmin}=1.00$

9 Scuffing load capacity

Results from AGMA 925-A03, Details see in the specific calculation sheet

Probability of wear (%)	[Pwear]	93.387
Probability of wear acc. to Dowson and Toyoda		
Probability of scuffing (%)	[Pscuff]	5% or lower

10 Measurements for tooth thickness

10.1 Tooth thickness tolerances

		----- Gear 1 -----	Gear 2 -----
Tooth thickness tolerance		DIN 3967 cd25	DIN 3967 cd25
Tooth thickness allowance (normal section) (in)	[As.e/i]	-0.0028/-0.0043	-0.0028/-0.0043

10.2 Base tangent lengths

		----- Gear 1 -----	Gear 2 -----
Number of teeth spanned	[k]	4.000	6.000
Base tangent length (no backlash) (in)	[Wk]	1.0672	1.6547
Base tangent length with allowance (in)	[Wk.e/i]	1.0647/1.0632	1.6522/1.6507
(in)	[ΔWk.e/i]	-0.0025/-0.0040	-0.0025/-0.0040
Diameter of measuring circle (in)	[dMWk.m]	2.4587	3.7952

10.3 Measurement over balls and pins

		----- Gear 1 -----	Gear 2 -----
Theoretical diameter of ball/pin (in)	[DM]	0.1794	0.1716
Effective diameter of ball/pin (in)	[DMeff]	0.1920	0.1728
Radial single-ball measurement backlash free (in)	[MrK]	1.3791	1.9925
Radial single-ball measurement (in)	[MrK.e/i]	1.3765/1.3750	1.9895/1.9878
Diameter of measuring circle (in)	[dMMr.m]	2.4719	3.7344
Diametral measurement over two balls without clearance (in)	[MdK]	2.7582	3.9816
Diametral measurement over two balls (in)	[MdK.e/i]	2.7530/2.7500	3.9756/3.9722
Diametral measurement over pins without clearance (in)	[MdR]	2.7582	3.9850
Measurement over pins according to DIN 3960 (in)	[MdR.e/i]	2.7530/2.7500	3.9790/3.9756
Measurement over 2 pins, free, AGMA 2002 (in)	[dk2f.e/i]	0.0000/0.0000	3.9754/3.9720
Measurement over 2 pins, transverse, AGMA 2002 (in)	[dk2t.e/i]	0.0000/0.0000	3.9824/3.9789
Measurement over 3 pins, axial, AGMA 2002 (in)	[dk3A.e/i]	2.7530/2.7500	3.9790/3.9756

10.4 Tooth thickness

		----- Gear 1 -----	Gear 2 -----
Reference chordal height from da.m (in)	[ha]	0.1219	0.1055
Chordal tooth thickness (in)	[sc]	0.1723	0.1599
(in)	[sc.e/i]	0.1696/0.1681	0.1572/0.1557
Tooth thickness, arc (in)	[sn]	0.1725	0.1600
(in)	[sn.e/i]	0.1697/0.1681	0.1572/0.1557

10.5 Backlash

		----- Pair -----	
Circumferential backlash from Aa (in)	[jtw_Aa.e/i]	0.0005/-0.0005	
Radial backlash (in)	[jrw.e/i]	0.0107/0.0058	
Circumferential backlash (transverse section) (in)	[jtw.e/i]	0.0095/0.0052	
Normal backlash (in)	[jn.e/i]	0.0085/0.0046	
Center distance allowances (in)	[Aa.e/i]	0.0006/-0.0006	
Backlash free center distance (in)	[aControl.e/i]	3.0927/3.0890	
Backlash free center distance, allowances (in)	[jta]	-0.0064/-0.0101	
		----- Gear 1 ----- Gear 2 -----	
dNf.i with aControl (in)	[dNf0.i]	2.2943	3.5838
Reserve (dNf0.i-dFf.e)/2 (in)	[cF0.i]	0.0040	0.0108
Tip clearance (in)	[c0.i(aControl)]	0.0179	0.0179
Torsional angle on input with output fixed:			
Total torsional angle (°)	[j.tSys]	0.4446/0.2423	

11 Tothing tolerances

		----- Gear 1 -----	Gear 2 -----
Following AGMA 2000-A88			
Accuracy grade	[Q-AGMA2000]	10	10
Pitch Variation Allowable (mil)	[VpA]	0.362	0.394
Runout Radial Tolerance (mil)	[VrT]	1.181	1.339
Profile Tolerance (mil)	[VphiT]	0.472	0.512
Tooth Alignment Tolerance (mil)	[VpsiT]	0.433	0.433
Composite Tolerance, Tooth-to-Tooth (mil)	[VqT]	0.748	0.748
Composite Tolerance, Total (mil)	[VcqT]	1.929	2.087
AGMA <-> ISO: VpA <-> fpbT, VrT <-> FrT, VpsiT <-> FbT, VqT <-> fidT, VcqT <-> FidT			
According to AGMA 2015-1-A01 & 2015-2-A06			
Accuracy grade	[Q-AGMA2015]	A7	A7
Single pitch deviation (mil)	[fptT]	0.472	0.472
Total cumulative pitch deviation (mil)	[FpT]	1.772	1.850
Profile form deviation (mil)	[ffaT]	0.472	0.472
Profile slope deviation (mil)	[fHaT]	0.374	0.394
Total profile deviation (mil)	[FaT]	0.591	0.630
Helix form deviation (mil)	[ffbT]	0.472	0.512
Helix slope deviation (mil)	[fHbT]	0.472	0.512
Total helix deviation (mil)	[FbT]	0.669	0.709
Single flank composite, total (mil)	[FisT]	1.969	2.047
Single flank composite, tooth-to-tooth (mil)	[fisT]	0.177	0.185
Radial composite, total (mil)	[FidT]	1.811	1.890
Radial composite, tooth-to-tooth (mil)	[fidT]	0.335	0.354

According to AGMA 2015-2-B15

Radial composite, total (mil)	[FidT]	2.835(R25)	2.87 (R25)
Radial composite, tooth-to-tooth (mil)	[fidT]	1.575(R25)	1.50 (R25)

Axis alignment tolerances recommendation acc. to ISO TR 10064-3:1996, Quality 7

Maximum value for deviation error of axis (mil)	[fΣβ]	0.793	(F _β =20.00)
Maximum value for inclination error of axes (mil)	[fΣδ]	1.586	

12 Modifications and determination of the tooth form

12.1 Data for the tooth form calculation

Data not available.

Please run the calculation in the "Tooth form" tab and open the main report again.

13 Supplementary data

Maximal possible center distance (eps_a=1.0)	[aMAX]	80.125
--	--------	--------

13.1 Masses, stiffnesses and moments of inertia

		----- Gear 1 -----	Gear 2 -----
Mass (lb)	[m]	1.831	4.229
Total mass (lb)	[mGes]	6.060	
Moment of inertia for system, relative to the input: calculation without consideration of the exact tooth shape			
Single gears, (da+df)/2...di (lb*ft²)	[J]	0.009422	0.05065
System (da+df)/2...di (lb*ft²)	[J]	0.03073	
Singular tooth stiffness (lbf/in/μin)	[c']	2.037	
Meshing stiffness (lbf/in/μin)	[cV]	2.720	

13.2 Wear, power loss, sound pressure level

Torsional stiffness at driving gear with fixed driven gear:

Torsional stiffness (1E6*lb/rad)	[cr]	0.326
Torsion when subjected to nominal torque (°)	[δcr]	0.020
Average coefficient of friction according to Niemann	[μm]	0.098
Wear sliding coef. by Niemann	[ζw]	0.745
Loss factor	[HV]	0.133
Gear power loss (hp)	[PVZ]	0.260
Meshing efficiency (%)	[ηz]	98.695
Sound pressure level based on Masuda, without PPTE/δs	[dB(A)]	62.467

14 Service life, damage

Required safety for tooth root	[S _{Fmin}]	1.00
Required safety for tooth flank	[S _{Hmin}]	1.00
Required service life	[H]	15000.00

Service life (calculated with required safeties):

System service life (h)	[H _{att}]	> 1000000
-------------------------	---------------------	-----------

		----- Gear 1 -----	Gear 2 -----
Tooth root service life (h)	[H _{Fatt}]	1e+06	1e+06
Tooth flank service life (h)	[H _{Hatt}]	1e+06	1e+06

Note: The entry 1e+006 h means that the Service life > 1,000,000 h.

14.1 Damage

Damage relative to the required service life (H, 15000.0 h)

F ₁ (%)	F ₂ (%)	H ₁ (%)	H ₂ (%)
0.0000	0.0000	0.0007	0.0004

15 Remarks

15.1 Conventions

- Specifications with **.e/i** mean: Maximum value **.e** and Minimum value **.i**, taking all tolerances into account.
- Specifications with **.m** mean: Mean value within tolerance.
- The center distance tolerances and the tooth thickness allowance are taken into account for the backlash tolerance. The maximum and minimum clearance corresponding to the largest and smallest allowances are shown. The calculation is performed for the operating pitch circle.
- Symbols used in **[]**: [xx,yy] xx as used in AGMA 2001-D04, yy as used in AGMA 2101-D04

15.2 Calculations and factors

- $s_{ateff} = s_{at} \cdot K_L / K_T / K_R \cdot K_{wb} / S_F$ ($S_F = 1.0$)
- LACR = Spur gear or helical gear with $\varepsilon_\beta (m_F) < 1.0$
- PSTC = Point of Single Tooth Contact
- Calculation of Z_β acc. to Corrigendum 1 ISO 6336-2:2008 with $Z_\beta = 1 / (\cos(\beta))^{0.5}$

_O.DefaultGeometry.GearBox.Shaft1.Shaft1_calc

1 Input data

Coordinate system shaft: see picture W-002

1.1 Shafts

1.1.1 Shaft 'Shaft1'

Drawing	
Initial position (in)	0.0000
Length (in)	0.0000
Speed (1/min)	1500.000
Direction of rotation:	clockwise
Material	C45 (1)
Young's modulus (lbf/in²)	29877900.100
Poisson's ratio nu	0.300
Density (lb/ft³)	488.8112
Coefficient of thermal expansion (10^-6/K)	11.500
Temperature (°F)	68.000
Temperature for load spectrum	
No.	Temperature (°C)
1	20.000
2	20.000
3	20.000
4	20.000
5	20.000
6	20.000
7	20.000
8	20.000
9	20.000
10	20.000
Weight of shaft (oz)	0.0000
Weight of shaft, including additional masses (oz)	0.0000
Mass moment of inertia (lb*in²)	0.0000
Momentum of mass GD2 (lb*ft²)	0.000

1.2 Weight force

The direction of the weight is not considered

1.3 Shaft modeling

1.4 Shear deformations

Consider deformations due to shearing

Shear correction factor 1.100

1.5 Rolling bearings

Contact angle of rolling bearings is considered

Tolerance field: Mean value

1.6 Housing

Reference temperature (°F) 68.000

1.7 Load applications

2 Shaft definition (Shaft1)

2.1 Outer contour

2.1.1 Cylinder (Cylinder), 0.0000in...2.8000in

Diameter (in)	[d]	0.86614
Length (in)	[l]	2.80000
Surface roughness (mil)	[Rz]	0.31496

2.2 Forces

2.2.1 Coupling (Coupling1(Input))

Position on shaft (in)	[y _{local}]	2.60000
Position in global system (in)	[y _{global}]	2.60000
Effective diameter (in)		1.18110
Radial force factor (-)		0.0000
Direction of the radial force (°)		0.0000
Axial force factor (-)		0.0000
Length of load application (in)		0.39370
Power (hp)		20.0000
Torque (ft*lb)		70.0277
Mass (lb)		0.0000
Mass moment of inertia J _p (lb*in ²)		0.0000e+00
Mass moment of inertia J _{xx} (lb*in ²)		0.0000e+00
Mass moment of inertia J _{zz} (lb*in ²)		0.0000e+00
Eccentricity (in)		0.00000

Load spectrum, driven (input)

Example with file (basic)

No.	Frequency(%)	Speed (1/min)	Power (hp)	Torque (ft*lb)

1	10.0000	1500.0000	19.9996	70.0277
2	10.0000	1500.0000	17.9996	63.0250
3	10.0000	1500.0000	15.9997	56.0222
4	10.0000	1500.0000	13.9997	49.0194
5	10.0000	1500.0000	11.9997	42.0166
6	10.0000	1500.0000	9.9998	35.0139
7	10.0000	1500.0000	7.9998	28.0111
8	10.0000	1500.0000	5.9999	21.0083
9	10.0000	1500.0000	3.9999	14.0055
10	10.0000	1500.0000	2.0000	7.0028

No.	Axial force (lbf)	Shearing force X (lbf)	Shearing force Z (lbf)
1	0.0000	0.0000	0.0000
2	0.0000	0.0000	0.0000
3	0.0000	0.0000	0.0000
4	0.0000	0.0000	0.0000
5	0.0000	0.0000	0.0000
6	0.0000	0.0000	0.0000
7	0.0000	0.0000	0.0000
8	0.0000	0.0000	0.0000
9	0.0000	0.0000	0.0000
10	0.0000	0.0000	0.0000

2.2.2 Power loss (Losses_z1)

Position on shaft (in)	[y _{local}]	1.20000
Position in global system (in)	[y _{global}]	1.20000
Power loss (hp)		0.0020
equivalent torque (ft*lbf)		-0.0072
Length of load application (in)		1.10000

Load spectrum, driving (output)

Example with file (basic)

No.	Frequency(%)	Speed (1/min)	Power (hp)	Torque (ft*lbf)
1	10.0000	1500.0000	-0.0020	-0.0072
2	10.0000	1500.0000	-0.0018	-0.0065
3	10.0000	1500.0000	-0.0016	-0.0057
4	10.0000	1500.0000	-0.0014	-0.0050
5	10.0000	1500.0000	-0.0012	-0.0043
6	10.0000	1500.0000	-0.0010	-0.0036
7	10.0000	1500.0000	-0.0008	-0.0029
8	10.0000	1500.0000	-0.0006	-0.0022
9	10.0000	1500.0000	-0.0004	-0.0014

10	10.0000	1500.0000	-0.0002	-0.0007
----	---------	-----------	---------	---------

2.2.3 Cylindrical gear (z1)

Position on shaft (in)	[y _{local}]	1.20000
Position in global system (in)	[y _{global}]	1.20000
Operating pitch diameter (in)		2.33448
Helix angle (°)		12.0000 Helix left hand
Working pressure angle at normal section (°)		22.5000
Position of contact (°)		-90.0000
Length of load application (in)		1.10000
Power (hp)		19.9981
Torque (ft*lb)		-70.0209

Load spectrum, driving (output)

Example with file (basic)

No.	Frequency(%)	Speed (1/min)	Power (hp)	Torque (ft*lb)
1	10.0000	1500.0000	-19.9976	-70.0209
2	10.0000	1500.0000	-17.9979	-63.0188
3	10.0000	1500.0000	-15.9981	-56.0167
4	10.0000	1500.0000	-13.9983	-49.0146
5	10.0000	1500.0000	-11.9986	-42.0125
6	10.0000	1500.0000	-9.9988	-35.0105
7	10.0000	1500.0000	-7.9991	-28.0084
8	10.0000	1500.0000	-5.9993	-21.0063
9	10.0000	1500.0000	-3.9995	-14.0042
10	10.0000	1500.0000	-1.9998	-7.0021

No.	Axial force (lbf)	Shearing force X (lbf)	Shearing force Z (lbf)	Bending moment X (ft*lb)	Bending moment Z (ft*lb)
1	-153.0116	719.8630	304.8385	-14.8834	-0.0000
2	-137.7105	647.8767	274.3546	-13.3951	-0.0000
3	-122.4093	575.8904	243.8708	-11.9067	-0.0000
4	-107.1081	503.9041	213.3869	-10.4184	-0.0000
5	-91.8070	431.9178	182.9031	-8.9300	-0.0000
6	-76.5058	359.9315	152.4192	-7.4417	-0.0000
7	-61.2046	287.9452	121.9354	-5.9534	-0.0000
8	-45.9035	215.9589	91.4515	-4.4650	-0.0000
9	-30.6023	143.9726	60.9677	-2.9767	-0.0000
10	-15.3012	71.9863	30.4838	-1.4883	-0.0000

3 Results

Note: the maximum deflection and torsion of the shaft under torque, the life modification factor aISO, and the bearing's minimum lubricant film thickness EHL, are predefined for the first load bin.

3.1 Shafts

Maximum deflection (mil)	0.0000
Position of the maximum (in)	0.0000
Mass center of gravity (in)	1.5000
Total axial load (lbf)	-159.832
Torsion of the shaft under torque (°)	0.114
(Difference between left and right shaft end)	

3.2 Bearing

3.3 Damage relative to the required service life ($L_{req} = 15000$ h)

Load case	B1	B2
1	9999.9 9	9999.9 9
2	9999.9 9	9999.9 9
3	9999.9 9	9999.9 9
4	9999.9 9	9999.9 9
5	9999.9 9	9999.9 9
Σ	9999.9 9	9999.9 9

B1 : RollerBearing1 (TIMKEN JL22349F-JL22310)

B2 : RollerBearing2 ()

3.4 Utilization relative to the required service life ($L_{req} = 15000$ h)

Rolling bearings	B1	B2
Utilization (%)	51730. 78	53470. 93

Note: Utilization = $(L_{req}/L_h)^{(1/k)}$

Ball bearing: $k = 3$, roller bearing: $k = 10/3$

B1 : RollerBearing1 (TIMKEN JL22349F-JL22310)

B2 : RollerBearing2 ()

3.5 Calculation of the factors required to define reliability R(t) using the Weibull distribution. t in (h)

Reliability not calculated

Bearing	p	fac	b	t0	T	R(H)%
1	Bearing	0	0	0	0	0.00
2	Bearing	0	0	0	0	0.00

Reliability of the configuration for required service life (%) 99.00

_O.DefaultGeometry.GearBox.Shaft2.Shaft2_calc

1 Input data

Coordinate system shaft: see picture W-002

1.1 Shafts

1.1.1 Shaft 'Shaft2'

Drawing	
Initial position (in)	0.0000
Length (in)	5.0000
Speed (1/min)	906.250
Direction of rotation:	clockwise
Material	C45 (1)
Young's modulus (lbf/in²)	29877900.100
Poisson's ratio nu	0.300
Density (lb/ft³)	488.8112
Coefficient of thermal expansion (10^-6/K)	11.500
Temperature (°F)	68.000
Temperature for load spectrum	
No. Temperature (°C)	
1	20.000
2	20.000
3	20.000
4	20.000
5	20.000
6	20.000
7	20.000
8	20.000
9	20.000
10	20.000
Weight of shaft (lb)	1.736
Weight of shaft, including additional masses (lb)	1.736
Mass moment of inertia (lb*in²)	2.9031
Momentum of mass GD2 (lb*ft²)	0.009

1.2 Weight force

The direction of the weight is not considered

1.3 Shaft modeling

1.4 Shear deformations

Consider deformations due to shearing
Shear correction factor1.100

1.5 Rolling bearings

Contact angle of rolling bearings is considered
Tolerance field:Mean value

1.6 Housing

Reference temperature (°F)68.000

1.7 Load applications

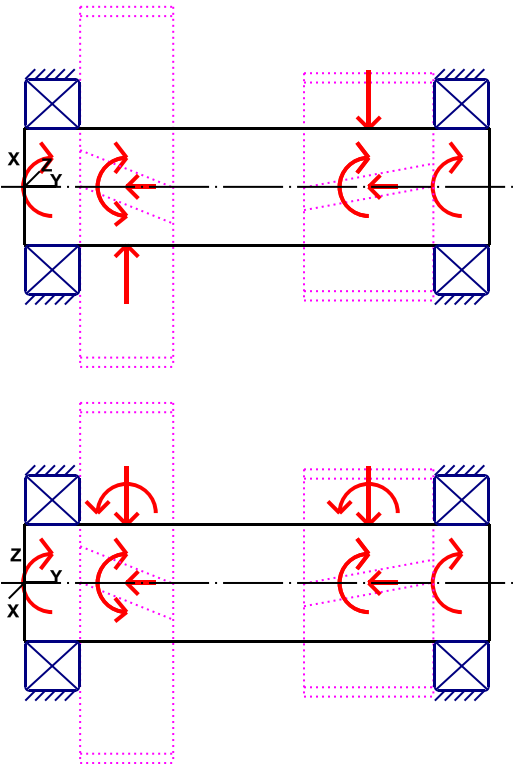


Figure: Load applications

2 Shaft definition (Shaft2)

2.1 Outer contour

2.1.1 Cylinder (Cylinder), 0.0000in...5.0000in

Diameter (in)	[d]	1.25000
Length (in)	[l]	5.00000
Surface roughness (mil)	[Rz]	0.31496

2.2 Forces

2.2.1 Power loss (Losses_RollerBearing3)

Position on shaft (in)	[y _{local}]	0.30000
Position in global system (in)	[y _{global}]	0.30000
Power loss (hp)		0.0473
equivalent torque (ft*lb)		-0.2743
Length of load application (in)		0.57811

Load spectrum, driving (output)

Example with file (basic)

No.	Frequency(%)	Speed (1/min)	Power (hp)	Torque (ft*lb)
1	10.0000	906.2500	-0.0473	-0.2743
2	10.0000	906.2500	-0.0426	-0.2469
3	10.0000	906.2500	-0.0379	-0.2194
4	10.0000	906.2500	-0.0331	-0.1920
5	10.0000	906.2500	-0.0284	-0.1646
6	10.0000	906.2500	-0.0237	-0.1372
7	10.0000	906.2500	-0.0189	-0.1097
8	10.0000	906.2500	-0.0142	-0.0823
9	10.0000	906.2500	-0.0095	-0.0549
10	10.0000	906.2500	-0.0047	-0.0274

2.2.2 Power loss (Losses_RollerBearing4)

Position on shaft (in)	[y _{local}]	4.70000
Position in global system (in)	[y _{global}]	4.70000
Power loss (hp)		0.0387
equivalent torque (ft*lb)		-0.2241
Length of load application (in)		0.57811

Load spectrum, driving (output)

Example with file (basic)

No.	Frequency(%)	Speed (1/min)	Power (hp)	Torque (ft*lb)
1	10.0000	906.2500	-0.0387	-0.2241
2	10.0000	906.2500	-0.0348	-0.2017
3	10.0000	906.2500	-0.0309	-0.1793
4	10.0000	906.2500	-0.0271	-0.1568
5	10.0000	906.2500	-0.0232	-0.1344
6	10.0000	906.2500	-0.0193	-0.1120
7	10.0000	906.2500	-0.0155	-0.0896
8	10.0000	906.2500	-0.0116	-0.0672
9	10.0000	906.2500	-0.0077	-0.0448

10	10.0000	906.2500	-0.0039	-0.0224
----	---------	----------	---------	---------

2.2.3 Power loss (Losses_z2)

Position on shaft (in)	[y _{local}]	1.10000
Position in global system (in)	[y _{global}]	1.10000
Power loss (hp)		0.0166
equivalent torque (ft*lb)		-0.0962
Length of load application (in)		1.00000

Load spectrum, driving (output)

Example with file (basic)

No.	Frequency(%)	Speed (1/min)	Power (hp)	Torque (ft*lb)
1	10.0000	906.2500	-0.0166	-0.0962
2	10.0000	906.2500	-0.0149	-0.0866
3	10.0000	906.2500	-0.0133	-0.0770
4	10.0000	906.2500	-0.0116	-0.0674
5	10.0000	906.2500	-0.0100	-0.0577
6	10.0000	906.2500	-0.0083	-0.0481
7	10.0000	906.2500	-0.0066	-0.0385
8	10.0000	906.2500	-0.0050	-0.0289
9	10.0000	906.2500	-0.0033	-0.0192
10	10.0000	906.2500	-0.0017	-0.0096

2.2.4 Power loss (Losses_z3)

Position on shaft (in)	[y _{local}]	3.70000
Position in global system (in)	[y _{global}]	3.70000
Power loss (hp)		0.0034
equivalent torque (ft*lb)		-0.0199
Length of load application (in)		1.39000

Load spectrum, driving (output)

Example with file (basic)

No.	Frequency(%)	Speed (1/min)	Power (hp)	Torque (ft*lb)
1	10.0000	906.2500	-0.0034	-0.0199
2	10.0000	906.2500	-0.0031	-0.0179
3	10.0000	906.2500	-0.0027	-0.0159
4	10.0000	906.2500	-0.0024	-0.0139
5	10.0000	906.2500	-0.0021	-0.0119
6	10.0000	906.2500	-0.0017	-0.0099
7	10.0000	906.2500	-0.0014	-0.0079
8	10.0000	906.2500	-0.0010	-0.0060
9	10.0000	906.2500	-0.0007	-0.0040
10	10.0000	906.2500	-0.0003	-0.0020

2.2.5 Cylindrical gear (z2)

Position on shaft (in)	[y _{local}]	1.10000
Position in global system (in)	[y _{global}]	1.10000
Operating pitch diameter (in)		3.86396
Helix angle (°)		12.0000 Helix right hand
Working pressure angle at normal section (°)		22.5000
Position of contact (°)		90.0000
Length of load application (in)		1.00000
Power (hp)		19.9981
Torque (ft*lb)		115.8967

Load spectrum, driven (input)

Example with file (basic)

No.	Frequency(%)	Speed (1/min)	Power (hp)	Torque (ft*lb)
1	10.0000	906.2500	19.9976	115.8967
2	10.0000	906.2500	17.9979	104.3070
3	10.0000	906.2500	15.9981	92.7173
4	10.0000	906.2500	13.9983	81.1277
5	10.0000	906.2500	11.9986	69.5380
6	10.0000	906.2500	9.9988	57.9483
7	10.0000	906.2500	7.9991	46.3587
8	10.0000	906.2500	5.9993	34.7690
9	10.0000	906.2500	3.9995	23.1793
10	10.0000	906.2500	1.9998	11.5897

No.	Axial force (lbf)	Shearing force X (lbf)	Shearing force Z (lbf)	Bending moment X (ft*lb)	Bending moment Z (ft*lb)
1	-153.0116	719.8630	-304.8385	24.6346	-0.0000
2	-137.7105	647.8767	-274.3546	22.1711	-0.0000
3	-122.4093	575.8904	-243.8708	19.7077	-0.0000
4	-107.1081	503.9041	-213.3869	17.2442	-0.0000
5	-91.8070	431.9178	-182.9031	14.7808	-0.0000
6	-76.5058	359.9315	-152.4192	12.3173	-0.0000
7	-61.2046	287.9452	-121.9354	9.8538	-0.0000
8	-45.9035	215.9589	-91.4515	7.3904	-0.0000
9	-30.6023	143.9726	-60.9677	4.9269	-0.0000
10	-15.3012	71.9863	-30.4838	2.4635	-0.0000

2.2.6 Cylindrical gear (z3)

Position on shaft (in)	[y _{local}]	3.70000
Position in global system (in)	[y _{global}]	3.70000
Operating pitch diameter (in)		2.43864
Helix angle (°)		12.1025 Helix left hand
Working pressure angle at normal section (°)		23.6279
Position of contact (°)		90.0000
Length of load application (in)		1.39000
Power (hp)		19.8920

Torque (ft*lb) -115.2822

Load spectrum, driving (output)

Example with file (basic)

No.	Frequency(%)	Speed (1/min)	Power (hp)	Torque (ft*lb)
1	10.0000	906.2500	-19.8916	-115.2822
2	10.0000	906.2500	-17.9024	-103.7540
3	10.0000	906.2500	-15.9133	-92.2258
4	10.0000	906.2500	-13.9241	-80.6975
5	10.0000	906.2500	-11.9350	-69.1693
6	10.0000	906.2500	-9.9458	-57.6411
7	10.0000	906.2500	-7.9566	-46.1129
8	10.0000	906.2500	-5.9675	-34.5847
9	10.0000	906.2500	-3.9783	-23.0564
10	10.0000	906.2500	-1.9892	-11.5282

No.	Axial force (lbf)	Shearing force X (lbf)	Shearing force Z (lbf)	Bending moment X (ft*lb)	Bending moment Z (ft*lb)
1	-243.2809	-1134.5597	-507.6165	24.7197	-0.0000
2	-218.9528	-1021.1037	-456.8548	22.2477	-0.0000
3	-194.6247	-907.6477	-406.0932	19.7758	-0.0000
4	-170.2966	-794.1918	-355.3315	17.3038	-0.0000
5	-145.9685	-680.7358	-304.5699	14.8318	-0.0000
6	-121.6405	-567.2798	-253.8082	12.3598	-0.0000
7	-97.3124	-453.8239	-203.0466	9.8879	-0.0000
8	-72.9843	-340.3679	-152.2849	7.4159	-0.0000
9	-48.6562	-226.9119	-101.5233	4.9439	-0.0000
10	-24.3281	-113.4560	-50.7616	2.4720	-0.0000

2.3 Bearing

2.3.1 RollerBearing3 (TIMKEN 08125-08231)

Bearing type

Taper roller bearing (single row)

Bearing position (in)	[y _{lokal}]	0.3000
Bearing position (in)	[y _{global}]	0.3000
Attachment of external ring	Set fixed bearing left	
Inner diameter (in)	[d]	1.2500
External diameter (in)	[D]	2.3125
Width (in)	[b]	0.5781
Corner radius (in)	[r]	0.0394

The bearing pressure angle will be considered in the calculation

Contact angle (°)	[α]	17.613
Position (center of pressure) (in)	[y _{D,lokal}]	0.538
Position (center of pressure) (in)	[y _{D,global}]	0.538

Basic static load rating (lbf)	[C ₀]	7868.315
Basic dynamic load rating (lbf)	[C]	7126.445
Fatigue load limit (lbf)	[C _u]	0.000
Values for approximated geometry:		
Basic dynamic load rating (lbf)	[C _{theo}]	0.000
Basic static load rating (lbf)	[C _{0theo}]	0.000
Correction factor Basic dynamic load rating		
	[f _c]	1.000
Correction factor Basic static load rating		
	[f _{c0}]	1.000

2.3.2 RollerBearing4 (TIMKEN 08125-08231)

Bearing type	Taper roller bearing (single row)	
Bearing position (in)	[y _{lokal}]	4.7000
Bearing position (in)	[y _{global}]	4.7000
Attachment of external ring	Set fixed bearing right	
Inner diameter (in)	[d]	1.2500
External diameter (in)	[D]	2.3125
Width (in)	[b]	0.5781
Corner radius (in)	[r]	0.0394

The bearing pressure angle will be considered in the calculation

Contact angle (°)	[α]	17.613
Position (center of pressure) (in)	[y _{D,lokal}]	4.462
Position (center of pressure) (in)	[y _{D,global}]	4.462
Basic static load rating (lbf)	[C ₀]	7868.315
Basic dynamic load rating (lbf)	[C]	7126.445
Fatigue load limit (lbf)	[C _u]	0.000
Values for approximated geometry:		
Basic dynamic load rating (lbf)	[C _{theo}]	0.000
Basic static load rating (lbf)	[C _{0theo}]	0.000
Correction factor Basic dynamic load rating		
	[f _c]	1.000
Correction factor Basic static load rating		
	[f _{c0}]	1.000

3 Gears

Gears are considered only as load application.

4 Results

Note: the maximum deflection and torsion of the shaft under torque, the life modification factor aISO, and the bearing's minimum lubricant film thickness EHL, are predefined for the first load bin.

4.1 Shafts

Maximum deflection (mil)	0.2614
Position of the maximum (in)	3.0050
Mass center of gravity (in)	2.5000
Total axial load (lbf)	-396.293
Torsion of the shaft under torque (°)	-0.075
(Difference between left and right shaft end)	

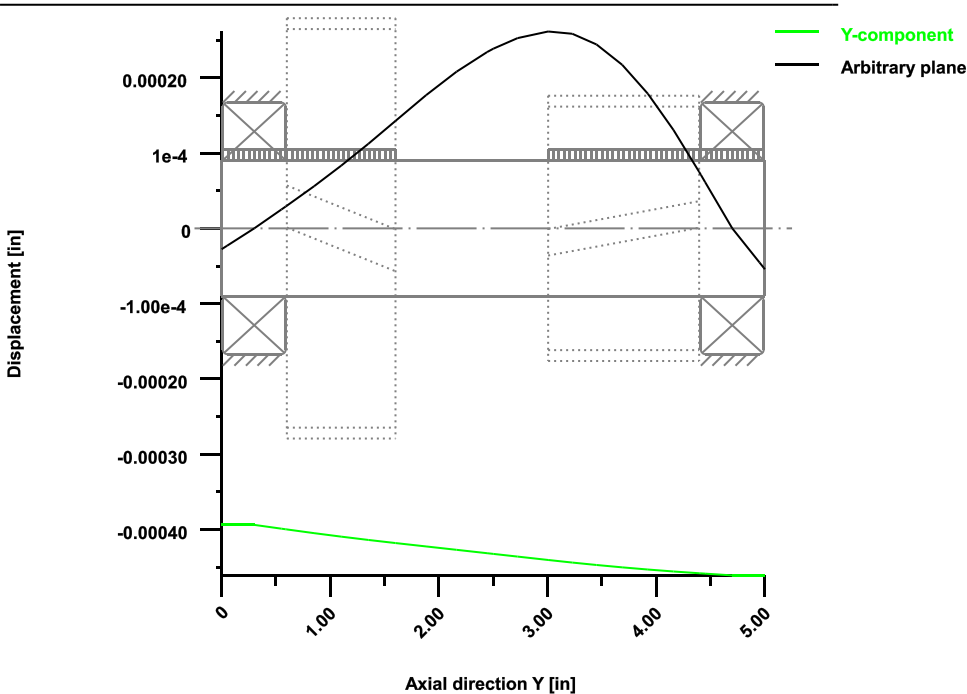
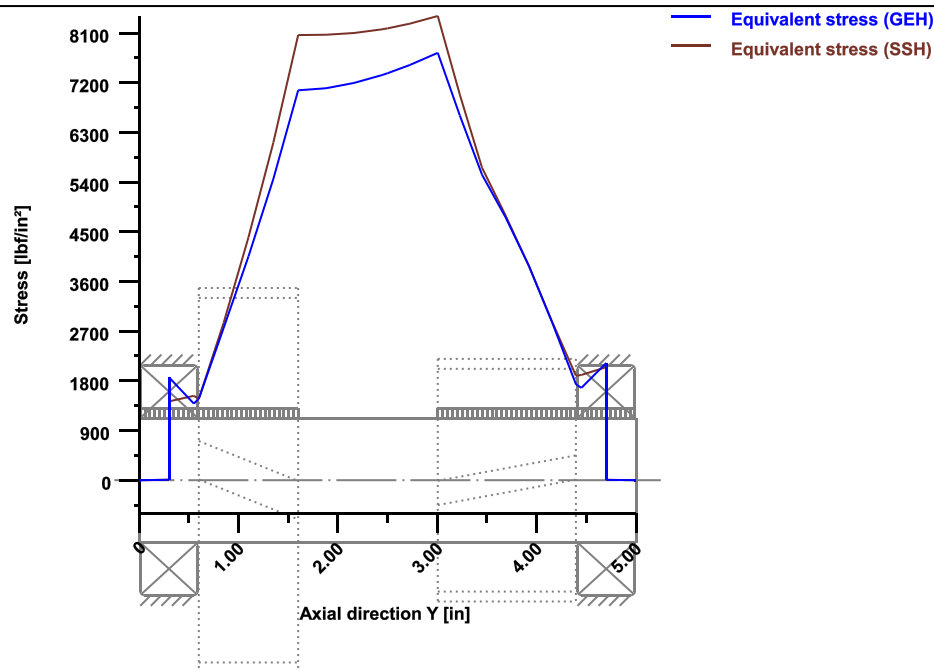


Figure: Deformation (bending etc.) (Arbitrary plane 226.411755 127)



Nominal stresses, without taking into account stress concentrations

GEH(von Mises): $\text{sigV} = ((\text{sigB} + \text{sigZ}, D)^2 + 3 * (\text{tauT} + \text{tauS})^2)^{1/2}$

SSH(Tresca): $\text{sigV} = ((\text{sigB} - \text{sigZ}, D)^2 + 4 * (\text{tauT} + \text{tauS})^2)^{1/2}$

Figure: Equivalent stress

4.2 Bearing

Probability of failure	[n]	10.00	%
Axial clearance (ISO 281)	[u _A]	0.39	mil
Lubricant	ISO-VG 220		
Lubricant - service temperature	[T _B]	158.00	°F
Rolling bearings, classical calculation (contact angle considered)			

4.3 'Shaft2' Rolling bearing 'RollerBearing3'

Position (Y-coordinate)	[y]	0.30	in
Life modification factor for reliability	[a ₁]	1.000	

4.3.1 Results according to ISO 281

Lubricant	ISO-VG 220		
Load ratio	[C/P]	5.988	
Operating viscosity	[v]	48.884	cSt
Reference viscosity	[v ₁]	0.000	cSt
Basic bearing rating life	[L _{nh}]	25290.52	h
Static safety factor	[S ₀]	9.36	

Bearing reaction force

Bearing reaction moment

	F _x (lbf)	F _y (lbf)	F _z (lbf)	F _r (lbf)	M _x (ft·lbf)	M _y (ft·lbf)	M _z (ft·lbf)	M _r (ft·lbf)
1	-396.436	739.345	510.622	646.449	10.122	0.000	7.858	12.814
2	-356.792	665.386	459.560	581.804	9.110	0.000	7.073	11.533
3	-317.148	591.427	408.497	517.159	8.098	0.000	6.287	10.252
4	-277.505	517.468	357.435	452.514	7.085	0.000	5.501	8.970
5	-237.861	443.508	306.373	387.869	6.073	0.000	4.715	7.689
6	-198.218	369.549	255.311	323.224	5.061	0.000	3.929	6.407
7	-158.574	295.590	204.249	258.579	4.049	0.000	3.143	5.126
8	-118.925	221.820	153.198	193.940	3.037	0.000	2.357	3.844
9	-79.283	147.852	102.132	129.293	2.025	0.000	1.572	2.563
10	-39.641	73.943	51.067	64.647	1.012	0.000	0.786	1.281

Displacement of bearing

Misalignment of bearing

	u _x (mil)	u _y (mil)	u _z (mil)	u _r (mil)	r _x (°)	r _y (°)	r _z (°)	r _r (°)
1	0.000	-0.394	0.000	0.000	-0.000	0.000	0.000	0.000
2	0.000	-0.394	0.000	0.000	-0.000	0.000	0.000	0.000
3	0.000	-0.394	0.000	0.000	-0.000	0.000	0.000	0.000
4	0.000	-0.394	0.000	0.000	-0.000	0.000	0.000	0.000
5	0.000	-0.394	0.000	0.000	-0.000	0.000	0.000	0.000
6	0.000	-0.394	0.000	0.000	-0.000	0.000	0.000	0.000
7	0.000	-0.394	0.000	0.000	-0.000	0.000	0.000	0.000
8	0.000	-0.394	-0.000	0.000	-0.000	0.000	0.000	0.000
9	0.000	-0.394	-0.000	0.000	-0.000	0.000	0.000	0.000
10	0.000	-0.394	-0.000	0.000	-0.000	0.000	0.000	0.000

4.4 'Shaft2' Rolling bearing 'RollerBearing4'

Position (Y-coordinate)	[y]	4.70	in
Life modification factor for reliability	[a ₁]	1.000	

4.4.1 Results according to ISO 281

Lubricant	ISO-VG 220		
Load ratio	[C/P]	8.235	
Operating viscosity	[v]	48.884	cSt
Reference viscosity	[v ₁]	0.000	cSt
Basic bearing rating life	[L _{nh}]	73158.06	h
Static safety factor	[S ₀]	9.09	

Bearing reaction force

Bearing reaction moment

	F _x (lbf)	F _y (lbf)	F _z (lbf)	F _r (lbf)	M _x (ft·lbf)	M _y (ft·lbf)	M _z (ft·lbf)	M _r (ft·lbf)
1	811.038	-343.325	301.716	865.341	-5.981	0.000	16.077	17.154

2	729.934	-308.992	271.544	778.807	-5.383	0.000	14.469	15.438
3	648.830	-274.660	241.373	692.273	-4.785	0.000	12.862	13.723
4	567.726	-240.327	211.201	605.739	-4.187	0.000	11.254	12.007
5	486.623	-205.995	181.029	519.204	-3.589	0.000	9.646	10.292
6	405.519	-171.662	150.858	432.670	-2.990	0.000	8.039	8.577
7	324.415	-137.330	120.686	346.136	-2.392	0.000	6.431	6.861
8	243.324	-103.017	90.527	259.618	-1.795	0.000	4.823	5.146
9	162.216	-68.678	60.351	173.079	-1.196	0.000	3.216	3.431
10	81.109	-34.341	30.177	86.541	-0.598	0.000	1.608	1.715

Displacement of bearing

Misalignment of bearing

	u_x (mil)	u_y (mil)	u_z (mil)	u_r (mil)	r_x (°)	r_y (°)	r_z (°)	r_r (°)
1	-0.000	-0.461	-0.000	0.000	0.000	-0.000	-0.000	0.000
2	-0.000	-0.454	-0.000	0.000	0.000	-0.000	-0.000	0.000
3	-0.000	-0.447	0.000	0.000	0.000	-0.000	-0.000	0.000
4	-0.000	-0.441	0.000	0.000	0.000	-0.000	-0.000	0.000
5	-0.000	-0.434	-0.000	0.000	0.000	-0.000	-0.000	0.000
6	-0.000	-0.427	-0.000	0.000	0.000	-0.000	-0.000	0.000
7	-0.000	-0.421	0.000	0.000	0.000	-0.000	-0.000	0.000
8	0.000	-0.414	0.000	0.000	0.000	-0.000	-0.000	0.000
9	-0.000	-0.407	-0.000	0.000	0.000	-0.000	-0.000	0.000
10	-0.000	-0.400	0.000	0.000	0.000	-0.000	-0.000	0.000

4.5 Damage relative to the required service life ($L_{req} = 15000$ h)

Load case	B1	B2
1	20.92	7.23
2	14.72	5.09
3	9.94	3.44
4	6.37	2.20
5	3.81	1.32
6	2.07	0.72
7	0.99	0.34
8	0.38	0.13
9	0.10	0.03
10	0.01	0.00
Σ	59.31	20.50

B1 : RollerBearing3 (TIMKEN 08125-08231)

B2 : RollerBearing4 (TIMKEN 08125-08231)

4.6 Utilization relative to the required service life ($L_{req} = 15000$ h)

Rolling bearings	B1	B2
Utilization (%)	85.49	62.17

Note: Utilization = $(L_{req}/L_h)^{(1/k)}$

Ball bearing: $k = 3$, roller bearing: $k = 10/3$

B1 : RollerBearing3 (TIMKEN 08125-08231)

B2 : RollerBearing4 (TIMKEN 08125-08231)

4.7 Calculation of the factors required to define reliability $R(t)$ using the Weibull distribution. t in (h)

Reliability not calculated

5 Strength calculation according to DIN 743:2012

5.1 Summary

Shaft2

Material	C45 (1)
Material type	Through hardened steel
Material treatment	unalloyed, through hardened
Surface treatment	No

Calculation of endurance limit and the static strength

Calculation for load case 2 ($\sigma_{av}/\sigma_{mv} = \text{const}$)

Cross section	Position (Y-Coord) (in)	
A-A	3.45	Smooth shaft

Results:

Cross section	$\beta\sigma$	$KF\sigma$	$K2d$	SD	SS
A-A	1.00	0.90	0.90	7.00	9.00

Required safeties:	1.20	1.20
--------------------	------	------

Abbreviations:

$\beta\sigma$: Notch factor, bending

$KF\sigma$: Surface factor

$K2d$: size factor bending

SD: Safety endurance limit

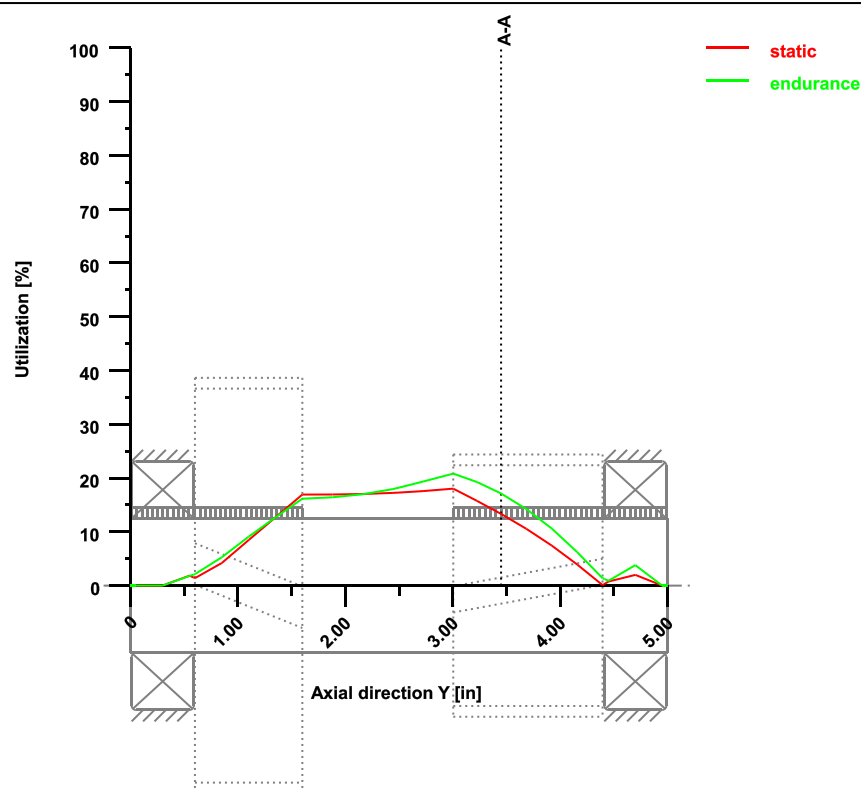
SS: Safety against yield point

5.2 Service life and damage

System service life (h)	[Hatt]	1000000.00
Damage to system (%)	[D]	0.00

5.2.1 Utilization (%)

Cross section	Static (yield point)	Endurance
A-A	13.336	17.133
Maximum utilization (%)	[A]	17.133



Utilization = S_{min}/S (%)

Figure: Strength (nominal load)

5.3 Calculation details

General statements

Label	Shaft2
Drawing	
Length (in)	[l] 5.000
lcElem =0	
Speed (1/min)	[n] 906.25

Material	C45 (1)
Material type	Through hardened steel
Material treatment	unalloyed, through hardened
Surface treatment	No

		Tension/Compression	Bending	Torsion	Shearing
Load factor static calculation		1.700	1.700	1.700	1.700
Load factor endurance limit		1.000	1.000	1.000	1.000
Reference diameter material (in)	[dB]			0.630	
σ_B according to DIN 743 (at dB) (lbf/in²)	[σB]			101526.85	
σ_B according to DIN 743 (at dB) (lbf/in²)	[σS]			71068.79	
[σdW], bei dB (lbf/in²)				40610.74	
[σbW], bei dB (lbf/in²)				50763.42	

[rtW], bei dB (lbf/in ²)	30458.05
Thickness of raw material (in) [dWerkst]	1.378
Material data calculated with K1(d), according to DIN 743/3	
Material strength calculated from size of raw material	
Geometric size factor K1d calculated from raw material diameter	
[σBeff] (lbf/in ²)	92553.24
[σSeff] (lbf/in ²)	62854.49
[σbFK] (lbf/in ²)	75425.39
[rtFK] (lbf/in ²)	43546.87
[σbBRand] (lbf/in ²)	91084.08
[σzdW] (lbf/in ²)	37021.30
[σbW] (lbf/in ²)	46276.62
[rtW] (lbf/in ²)	27765.97

Endurance limit for single stage use

Calculation for load case 2 ($\sigma_{av}/\sigma_{mv} = \text{const}$)

5.3.1 Cross section 'A-A' Smooth shaft

Comment

Position (Y-Coordinate) (in)	[y]	3.4501
External diameter (in)	[da]	1.2500
Inner diameter (in)	[di]	0.0000
Notch effect	Smooth shaft	
Mean roughness (mil)	[Rz]	0.3150

Tension/Compression Bending Torsion Shearing

Tension/Compression Bending Torsion Shearing

Load: (lbf) (ft*lbf)

Mean value	-254.3	0.0	-39.3	0.0
Amplitude	254.3	47.6	39.3	58.9
Maximum value	-864.8	81.0	133.6	100.0
Cross section, moment of resistance: (in ²) (in ³)				
[A, Wb, Wt, A]	1.2	0.2	0.4	1.2

Stresses: (lbf/in²)

[σzdm, σbm, τm, τqm] (lbf/in ²)	-207.262	0.000	-1229.764	0.000
[σzda, σba, τa, τqa] (lbf/in ²)	207.262	2980.513	1229.764	63.944
[σzdmax, σbmax, τmax, τqmax] (lbf/in ²)	-704.691	5066.872	4181.197	108.704

Technological size influence	[K1(σB)]	0.912
	[K1(σS)]	0.884

Tension/Compression Bending Torsion

Notch effect coefficient	[β]	1.000	1.000	1.000
Geometrical size influence	[K2(d)]	1.000	0.904	0.904
Influence coefficient surface roughness	[KF]	0.900	0.900	0.942
Surface stabilization factor	[KV]	1.000	1.000	1.000
Total influence coefficient	[K]	1.111	1.218	1.168

Present safety for endurance limit:

Equivalent mean stress (lbf/in ²)	[σmV]	2119.905
Equivalent mean stress (lbf/in ²)	[τmV]	1223.928

Fatigue limit of part (lbf/in ²)	[σWK]	33315.075	37998.496	23778.631
Influence coefficient of mean stress sensitivity.				
	[ψσK]	0.219	0.258	0.147
Permissible amplitude (lbf/in ²)	[σADK]	5597.945	32100.925	20736.688
Safety against fatigue	[S]			7.004
Required safety against fatigue	[Smin]			1.200
Result (%)	[S/Smin]			583.7

Present safety

for proof against exceed of yield point:

Static notch sensitivity factor	[K2F]	1.000	1.200	1.200
Increase coefficient	[γF]	1.000	1.000	1.000
Yield stress of part (lbf/in ²)	[σFK]	62854.494	75425.393	43546.871
Safety yield stress	[S]			8.998
Required safety	[Smin]			1.200
Result (%)	[S/Smin]			749.8

Remarks:

- The shearing force is not considered in the analysis specified in DIN 743.
- Cross section with interference fit: The notch factors for the 'Slight interference fit' case are no longer defined in DIN 743.
The notch factors are taken from the FKM Guideline, except if those for the 'tight interference fit' according to DIN are smaller, in which case they will be used.

_O.DefaultGeometry.GearBox.Shaft3.Shaft3_calc

1 Input data

Coordinate system shaft: see picture W-002

1.1 Shafts

1.1.1 Shaft 'Shaft3'

Drawing	
Initial position (in)	0.0000
Length (in)	0.0000
Speed (1/min)	587.838
Direction of rotation:	counterclockwise
Material	C45 (1)
Young's modulus (lbf/in²)	29877900.100
Poisson's ratio nu	0.300
Density (lb/ft³)	488.8112
Coefficient of thermal expansion (10^-6/K)	11.500
Temperature (°F)	68.000
Temperature for load spectrum	
No. Temperature (°C)	
1	20.000
2	20.000
3	20.000
4	20.000
5	20.000
6	20.000
7	20.000
8	20.000
9	20.000
10	20.000
Weight of shaft (oz)	0.0000
Weight of shaft, including additional masses (oz)	0.0000
Mass moment of inertia (lb*in²)	0.0000
Momentum of mass GD2 (lb*ft²)	0.000

1.2 Weight force

The direction of the weight is not considered

1.3 Shaft modeling

1.4 Shear deformations

Consider deformations due to shearing

Shear correction factor 1.100

1.5 Rolling bearings

Contact angle of rolling bearings is considered

Tolerance field: Mean value

1.6 Housing

Reference temperature (°F) 68.000

1.7 Load applications

2 Shaft definition (Shaft3)

2.1 Outer contour

2.1.1 Cylinder (Cylinder), 0.0000in...3.0000in

Diameter (in)	[d]	1.12520
Length (in)	[l]	3.00000
Surface roughness (mil)	[Rz]	0.31496

2.2 Forces

2.2.1 Coupling (Coupling2(Output))

Position on shaft (in)	[y _{local}]	2.80000
Position in global system (in)	[y _{global}]	2.80000
Effective diameter (in)		1.18110
Radial force factor (-)		0.0000
Direction of the radial force (°)		0.0000
Axial force factor (-)		0.0000
Length of load application (in)		0.39370
Power (hp)		19.8900
Torque (ft*lb)		177.7087
Mass (lb)		0.0000
Mass moment of inertia J _p (lb*in ²)		0.0000e+00
Mass moment of inertia J _{xx} (lb*in ²)		0.0000e+00
Mass moment of inertia J _{zz} (lb*in ²)		0.0000e+00
Eccentricity (in)		0.00000

Load spectrum, driving (output)

Example with file (basic)

No.	Frequency(%)	Speed (1/min)	Power (hp)	Torque (ft*lb)

1	10.0000	-587.8378	-19.8896	177.7087
2	10.0000	-587.8378	-17.9006	159.9378
3	10.0000	-587.8378	-15.9117	142.1669
4	10.0000	-587.8378	-13.9227	124.3961
5	10.0000	-587.8378	-11.9337	106.6252
6	10.0000	-587.8378	-9.9448	88.8543
7	10.0000	-587.8378	-7.9558	71.0835
8	10.0000	-587.8378	-5.9669	53.3126
9	10.0000	-587.8378	-3.9779	35.5417
10	10.0000	-587.8378	-1.9890	17.7709

No.	Axial force (lbf)	Shearing force X (lbf)	Shearing force Z (lbf)
1	0.0000	0.0000	0.0000
2	0.0000	0.0000	0.0000
3	0.0000	0.0000	0.0000
4	0.0000	0.0000	0.0000
5	0.0000	0.0000	0.0000
6	0.0000	0.0000	0.0000
7	0.0000	0.0000	0.0000
8	0.0000	0.0000	0.0000
9	0.0000	0.0000	0.0000
10	0.0000	0.0000	0.0000

2.2.2 Power loss (Losses_z4)

Position on shaft (in)	[y _{local}]	1.30000
Position in global system (in)	[y _{global}]	1.30000
Power loss (hp)		0.0020
equivalent torque (ft*lbf)		0.0179
Length of load application (in)		1.38000

Load spectrum, driving (output)

Example with file (basic)

No.	Frequency(%)	Speed (1/min)	Power (hp)	Torque (ft*lbf)
1	10.0000	-587.8378	-0.0020	0.0179
2	10.0000	-587.8378	-0.0018	0.0161
3	10.0000	-587.8378	-0.0016	0.0143
4	10.0000	-587.8378	-0.0014	0.0126
5	10.0000	-587.8378	-0.0012	0.0108
6	10.0000	-587.8378	-0.0010	0.0090
7	10.0000	-587.8378	-0.0008	0.0072
8	10.0000	-587.8378	-0.0006	0.0054
9	10.0000	-587.8378	-0.0004	0.0036

10	10.0000	-587.8378	-0.0002	0.0018
----	---------	-----------	---------	--------

2.2.3 Cylindrical gear (z4)

Position on shaft (in)	[y _{local}]	1.30000
Position in global system (in)	[y _{global}]	1.30000
Operating pitch diameter (in)		3.75956
Helix angle (°)		12.1025 Helix right hand
Working pressure angle at normal section (°)		23.6279
Position of contact (°)		-90.0000
Length of load application (in)		1.38000
Power (hp)		19.8920
Torque (ft*lb)		-177.7267

Load spectrum, driven (input)

Example with file (basic)

No.	Frequency(%)	Speed (1/min)	Power (hp)	Torque (ft*lb)
1	10.0000	-587.8378	19.8916	-177.7267
2	10.0000	-587.8378	17.9024	-159.9541
3	10.0000	-587.8378	15.9133	-142.1814
4	10.0000	-587.8378	13.9241	-124.4087
5	10.0000	-587.8378	11.9350	-106.6360
6	10.0000	-587.8378	9.9458	-88.8634
7	10.0000	-587.8378	7.9566	-71.0907
8	10.0000	-587.8378	5.9675	-53.3180
9	10.0000	-587.8378	3.9783	-35.5453
10	10.0000	-587.8378	1.9892	-17.7727

No.	Axial force (lbf)	Shearing force X (lbf)	Shearing force Z (lbf)	Bending moment X (ft*lb)	Bending moment Z (ft*lb)
1	243.2809	1134.5597	507.6165	38.1095	0.0000
2	218.9528	1021.1037	456.8548	34.2986	0.0000
3	194.6247	907.6477	406.0932	30.4876	0.0000
4	170.2966	794.1918	355.3315	26.6767	0.0000
5	145.9685	680.7358	304.5699	22.8657	0.0000
6	121.6405	567.2798	253.8082	19.0548	0.0000
7	97.3124	453.8239	203.0466	15.2438	0.0000
8	72.9843	340.3679	152.2849	11.4329	0.0000
9	48.6562	226.9119	101.5233	7.6219	0.0000
10	24.3281	113.4560	50.7616	3.8110	0.0000

3 Results

Note: the maximum deflection and torsion of the shaft under torque, the life modification factor aISO, and the bearing's minimum lubricant film thickness EHL, are predefined for the first load bin.

3.1 Shafts

Maximum deflection (mil)	0.0000
Position of the maximum (in)	0.0000
Mass center of gravity (in)	2.0000
Total axial load (lbf)	264.583
Torsion of the shaft under torque (°)	0.128
(Difference between left and right shaft end)	

3.2 Bearing

3.3 Damage relative to the required service life ($L_{req} = 15000$ h)

Load case	B1	B2
1	9999.9 9	9999.9 9
2	9999.9 9	9999.9 9
3	9999.9 9	9999.9 9
4	9999.9 9	9999.9 9
5	9999.9 9	9999.9 9
Σ	9999.9 9	9999.9 9

B1 : RollerBearing5 (TIMKEN L45449-L45410)

B2 : RollerBearing6 ()

3.4 Utilization relative to the required service life ($L_{req} = 15000$ h)

Rolling bearings	B1	B2
Utilization (%)	45168. 45	52002. 44

Note: Utilization = $(L_{req}/L_h)^{(1/k)}$

Ball bearing: $k = 3$, roller bearing: $k = 10/3$

B1 : RollerBearing5 (TIMKEN L45449-L45410)

B2 : RollerBearing6 ()

3.5 Calculation of the factors required to define reliability R(t) using the Weibull distribution. t in (h)

Reliability not calculated

Bearing	p	fac	b	t0	T	R(H)%
1	Bearing	0	0	0	0	0.00
2	Bearing	0	0	0	0	0.00

Reliability of the configuration for required service life (%) 99.00